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# **Practical Multilayer Network Analysis with Py3plex**

*Release 1.0.1*

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**Dec 07, 2025**



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## FRONT MATTER

### 1.1 About This Book

**Practical Multilayer Network Analysis with Py3plex** is a research-oriented handbook for graduate students and applied network scientists. It provides both theoretical foundations and practical guidance for analyzing complex systems using multilayer network representations.

This book covers:

- Mathematical foundations of multilayer networks
- The py3plex library architecture and design philosophy
- Hands-on examples and workflows for real network analysis
- **A powerful SQL-like DSL for network queries**
- Production deployment and reproducibility practices

#### DSL Feature Highlight

Py3plex includes a first-class DSL for querying networks with SQL-like syntax:

```
from py3plex.dsl import Q, L

# Find top influencers using builder API
result = (
    Q.nodes()
    .from_layers(L["social"])
    .where(degree__gt=5)
    .compute("betweenness centrality")
    .order_by("-betweenness centrality")
    .limit(10)
    .execute(network)
)
```

See **Part III** for complete DSL coverage!

### 1.2 Target Audience

This book is designed for:

- **Graduate students** in network science, computer science, and related fields
- **Applied researchers** working with complex relational data

- **Data scientists** analyzing social, biological, or infrastructure networks
- **Software engineers** building network analysis pipelines

### 1.2.1 Prerequisites

Readers should be familiar with:

- Python programming (intermediate level)
- Basic graph theory concepts
- Fundamental linear algebra

## 1.3 About the Software

**py3plex** (version 1.x) is an open-source Python library for multilayer and multiplex network analysis. It provides:

- Native support for multilayer network structures
- 17+ specialized algorithms for multilayer analysis
- A SQL-like query language (DSL) for network exploration
- NetworkX compatibility and interoperability
- High-performance I/O with Arrow/Parquet support

The library is actively maintained and tested on Python 3.8+.

## 1.4 How to Use This Book

**Part I** establishes foundations: what multilayer networks are, when to use them, and how py3plex is designed.

**Part II** teaches practical usage: installation, data loading, visualization, and running core algorithms.

**Part III** presents the DSL query language, a major feature for expressing complex analyses concisely.

**Part IV** demonstrates real-world applications through detailed case studies.

**Part V** covers systems topics: testing, reproducibility, and the web GUI.

**Appendices** provide reference material, detailed configurations, and extensive validation examples that supplement the main text.

### 1.4.1 Code Examples

All code examples in this book are runnable and tested. Examples use consistent naming and follow modern Python conventions. File paths reference the accompanying GitHub repository:

<https://github.com/SkBlaz/py3plex>

### 1.4.2 Conventions

- **Code blocks** use syntax highlighting and are self-contained
- **Mathematical notation** follows standard network science conventions
- **Feature status** is clearly marked:
  - **Stable** - Production-ready with guarantees
  - **Experimental** - Functional but may change

- **Planned** - Future features (mentioned only briefly)

## 1.5 Acknowledgments

This work builds on contributions from the py3plex community and research collaborations in multilayer network analysis.

If you use py3plex in your research, please cite:

```
@Article{Skrlj2019,
  author={Skrlj, Blaz and Kralj, Jan and Lavrac, Nada},
  title={Py3plex toolkit for visualization and analysis of multilayer networks},
  journal={Applied Network Science},
  year={2019},
  volume={4},
  number={1},
  pages={94},
  doi={10.1007/s41109-019-0203-7}
}
```

## 1.6 License

The py3plex library is released under the MIT License. This book’s content is licensed under Creative Commons Attribution 4.0 International (CC BY 4.0).

## 1.7 Version Information

- **Book version:** 1.0 (2025)
- **py3plex version:** 1.x
- **Python support:** 3.8+

### 1.7.1 Introduction & Motivation

#### Why Multilayer Networks?

Real-world systems rarely consist of a single type of relationship. Consider:

- **Social networks:** People interact through friendship, collaboration, family ties, and online connections—each relationship type has different meanings and dynamics.
- **Transportation systems:** Cities connect via air, rail, and road networks. A disruption in one mode affects the others, and travelers make choices across modes.
- **Biological systems:** Proteins interact through physical binding, regulatory relationships, and metabolic pathways—each layer represents a different mechanism.
- **Information systems:** Software components depend on each other through API calls, data flows, and deployment relationships.

Traditional single-layer (monoplex) networks flatten these diverse relationships into a single graph, losing critical structural information. **Multilayer networks** preserve the distinct nature of different relationship types by organizing them into separate but interconnected layers.

### The Cost of Flattening

When we flatten a multilayer network into a single graph, we lose:

1. **Relationship semantics:** A coauthor edge is fundamentally different from a Twitter follower edge, but flattening treats them identically.
2. **Community structure:** Groups that are well-separated in individual layers may appear spuriously connected when layers are combined.
3. **Node roles:** A person might be central in their professional network but peripheral in their hobby network—this context-dependent centrality disappears upon flattening.
4. **Cross-layer dynamics:** Information spreading from email to in-person meetings follows specific patterns that become invisible when layers are merged.

**Example:** A researcher with 2 coauthors and 500 Twitter followers has degree 502 in a flattened network, suggesting high influence. But their academic impact (degree 2) is actually modest—the Twitter followers don't write papers with them. Multilayer analysis preserves this distinction.

### What py3plex Enables

**py3plex** is a Python library designed for practical analysis of multilayer networks. It provides:

#### 1. Expressive Data Structures

Native support for multilayer network representations that mirror how real systems are naturally structured:

```
from py3plex.core import multinet

# Create a multilayer network
network = multinet.multi_layer_network()

# Add edges across multiple layers
network.add_edges([
    ['Alice', 'coauthor', 'Bob', 'coauthor', 1],
    ['Alice', 'twitter', 'Carol', 'twitter', 1],
    ['Bob', 'coauthor', 'Dave', 'coauthor', 1],
], input_type="list")
```

#### 2. Multilayer-Aware Algorithms

Algorithms specifically designed for multilayer structure:

- **Community detection** that respects layer boundaries while finding cross-layer modules
- **Centrality measures** that account for both intra-layer and inter-layer importance
- **Random walks** that can transition between layers
- **Dynamics models** (e.g., epidemic spreading) with layer-specific parameters

#### 3. A Query Language for Networks

A SQL-like DSL (Domain-Specific Language) for expressing complex network analyses concisely:

##### DSL Example: Concise Network Analysis

Instead of writing loops and conditions, use declarative queries:



```

from py3plex.dsl import Q, L

# Find high-degree nodes in the coauthor layer
result = (
    Q.nodes()
    .from_layers(L["coauthor"])
    .where(degree__gt=5)
    .compute("betweenness centrality", "pagerank")
    .order_by("-betweenness centrality")
    .limit(10)
    .execute(network)
)

```

```

# Export for further analysis
result.to_pandas().to_csv("top_researchers.csv")

```

This replaces dozens of lines of manual filtering and iteration with a clear, composable query.

#### Key DSL features:

- SQL-like syntax familiar to data analysts
- Layer algebra: `L["social"] + L["work"] - L["bots"]`
- Type-safe Python builder API
- Export to pandas, NetworkX, Arrow, CSV, JSON
- EXPLAIN mode for query optimization

## 4. Visualization for Multilayer Networks

Publication-ready visualizations that show layer structure and cross-layer connections:

```

# Visualize with layer separation
network.visualize_network(
    show=True,
    layout_style="force_directed_3d"
)

```

## 5. Production-Ready Infrastructure

- **High-performance I/O** with Arrow/Parquet support for large networks
- **NetworkX compatibility** for interoperability with the broader Python ecosystem
- **Testing and validation** frameworks to ensure correctness
- **Docker containers** and CLI tools for deployment

## Who Should Use This Book

This book is designed for:

### Graduate Students and Researchers

If you're analyzing social, biological, or infrastructure networks in your research, this book provides both theoretical foundations and practical tools. You'll learn when multilayer modeling is appropriate and how to apply specialized algorithms correctly.

### Data Scientists and Analysts

If you're working with complex relational data—user behavior across platforms, supply chain networks, knowledge graphs—py3plex offers a principled way to preserve and analyze structural nuances.

### Software Engineers

If you're building network analysis pipelines, this book shows how to use py3plex's APIs, DSL, and deployment tools to create maintainable, efficient systems.

### Prerequisites

This book assumes:

- **Intermediate Python** (classes, list comprehensions, basic NumPy)
- **Basic graph theory** (nodes, edges, paths, degree)
- **Fundamental linear algebra** (matrices, matrix multiplication)

You don't need prior multilayer network experience—Part I builds from fundamentals.

### Real-World Applications

Multilayer network analysis has proven valuable in numerous domains:

#### Biological Networks

- Protein-protein interactions across multiple tissues or conditions
- Gene regulatory networks with transcription, translation, and metabolic layers
- Brain connectivity with structural and functional layers

#### Social Networks

- User behavior across multiple social platforms
- Organizational networks with formal and informal relationships
- Citation networks with papers, authors, and venues

#### Infrastructure Networks

- Transportation with air, rail, and road connections
- Power grids with generation, transmission, and distribution layers
- Communication networks with physical and logical layers

#### Knowledge and Information

- Heterogeneous information networks (authors-papers-venues)
- Semantic networks with multiple relationship types
- Code dependency graphs with API, data, and deployment layers

Each of these applications benefits from multilayer analysis by capturing the true structure of the system rather than forcing it into a single-layer representation.

### How This Book Is Organized

#### Part I (Chapters 1-3): Foundations

Introduces multilayer networks, defines key concepts, and explains py3plex's design philosophy.

#### Part II (Chapters 4-7): Working with Py3plex

Covers installation, data loading, visualization, and core algorithms—everything needed for basic usage.

### Part III (Chapters 8-11): Advanced Analysis and the DSL

Deep dive into the query language, a major feature for expressing complex analyses concisely.

### Part IV (Chapters 12-14): Case Studies

Complete, reproducible analyses of real datasets across different domains.

### Part V (Chapters 15-17): Systems and Deployment

Testing, reproducibility, and production deployment including the web GUI.

### Appendices (A-E): Reference Material

Detailed configurations, validation scripts, error handling, and API reference.

Each chapter includes:

- **Conceptual explanations** with mathematical foundations where needed
- **Code examples** that are runnable and tested
- **Practical tips** based on real usage patterns
- **References** for deeper study

## Summary

Multilayer networks provide a natural and expressive way to model systems with multiple types of relationships. By preserving the distinct nature of different layers, we can:

- Analyze structure more accurately
- Compute meaningful centralities and communities
- Model dynamics more realistically
- Make better predictions and interventions

**py3plex** makes multilayer network analysis accessible and practical, with a focus on:

- Expressive data structures
- Specialized algorithms
- A powerful query language
- Production-ready tools

The following chapters will show you how to use these capabilities effectively.

## Further Reading

- Kivelä, M., et al. (2014). “Multilayer networks.” *Journal of Complex Networks* 2(3): 203-271. [Comprehensive review of multilayer network theory]
- Boccaletti, S., et al. (2014). “The structure and dynamics of multilayer networks.” *Physics Reports* 544(1): 1-122. [Physics-focused perspective]
- Bianconi, G. (2018). *Multilayer Networks: Structure and Function*. Oxford University Press. [Textbook treatment]

## 1.7.2 Multilayer Network Basics

This chapter establishes the formal foundations of multilayer networks. We define key concepts, introduce mathematical notation, and show how different types of multilayer networks relate to each other.

## Formal Definitions

### Node-Layer Pairs

A **multilayer network**  $\mathcal{M}$  consists of:

- A set of **nodes**  $V = \{v_1, v_2, \dots, v_N\}$
- A set of **layers**  $L = \{\alpha, \beta, \gamma, \dots\}$
- A set of **node-layer pairs**  $V_M = V \times L$
- **Intra-layer edges**  $E_\alpha \subseteq V \times V$  within each layer  $\alpha$
- **Inter-layer edges**  $E_C \subseteq V_M \times V_M$  connecting node-layer pairs

The **node-layer pair**  $(v, \alpha)$  is the fundamental unit: it represents node  $v$  in the context of layer  $\alpha$ .

### Supra-Adjacency Matrix

The **supra-adjacency matrix**  $\mathbf{A}$  is a block matrix that encodes the full multilayer structure:

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{\alpha\alpha} & \mathbf{A}_{\alpha\beta} & \cdots \\ \mathbf{A}_{\beta\alpha} & \mathbf{A}_{\beta\beta} & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

Where:

- **Diagonal blocks**  $\mathbf{A}_{\alpha\alpha}$  are the adjacency matrices of individual layers
- **Off-diagonal blocks**  $\mathbf{A}_{\alpha\beta}$  encode inter-layer connections

For a multiplex network with  $N$  nodes and  $L$  layers,  $\mathbf{A}$  is an  $(N \times L) \times (N \times L)$  matrix.

**Example:** A 3-node, 2-layer network:

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{\text{friends}} & \omega \mathbf{I} \\ \omega \mathbf{I} & \mathbf{A}_{\text{colleagues}} \end{pmatrix}$$

where  $\omega$  is the **inter-layer coupling strength** and  $\mathbf{I}$  is the identity matrix (representing identity edges: each node connects to itself across layers).

## Types of Multilayer Networks

### Multiplex Networks

**Definition:** A multiplex network has the same node set in all layers, with different edge sets per layer.

$$V_\alpha = V_\beta = \dots = V \quad \text{for all layers}$$

**Characteristics:**

- **Same entities** appear in all layers
- **Different relationship types** per layer
- **Strong inter-layer coupling** (typically  $\omega = 1.0$  for identity edges)

**Example:** Social Multiplex Network

```

from py3plex.core import multinet

# Create multiplex network
network = multinet.multi_layer_network(network_type="multiplex")

# Same people, different relationship types
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Bob', 'friends', 'Carol', 'friends', 1],
    ['Alice', 'colleagues', 'Bob', 'colleagues', 1],
    ['Bob', 'colleagues', 'Dave', 'colleagues', 1],
], input_type="list")

# Verify structure
print(f"Layers: {network.get_layers()}")
print(f"Nodes per layer: {network.get_number_of_nodes_per_layer()}")

```

**Real-world applications:**

- Social networks across platforms (Facebook, Twitter, LinkedIn)
- Transportation modes (air, rail, road)
- Communication channels (email, phone, chat)
- Biological interactions (genetic, protein, metabolic)

**Heterogeneous Information Networks**

**Definition:** A network with different node types, where edges connect specific node type pairs.

**Characteristics:**

- **Different node types** per layer (authors, papers, venues)
- **Type-specific edges** (author-paper, paper-venue)
- **No inter-layer coupling** (nodes don't repeat across layers)

**Example: Academic Network**

```

network = multinet.multi_layer_network()

# Different node types
network.add_edges([
    ['Alice', 'authors', 'Paper1', 'papers', 1],
    ['Bob', 'authors', 'Paper1', 'papers', 1],
    ['Paper1', 'papers', 'ICML', 'venues', 1],
    ['Paper2', 'papers', 'ICML', 'venues', 1],
], input_type="list")

```

This creates a **tripartite** network with three node types: authors, papers, and venues.

**Real-world applications:**

- Academic networks (authors, papers, venues, keywords)
- E-commerce (users, products, sellers, categories)
- Biomedical (drugs, diseases, targets, pathways)

- Knowledge graphs (entities, relations, concepts)

### Temporal Networks

**Definition:** Networks that evolve over time, represented as time-sliced layers.

**Characteristics:**

- **Time windows as layers** (2020, 2021, 2022, ...)
- **Node presence varies** across time slices
- **Temporal edges** connect adjacent time slices

**Example: Temporal Social Network**

```
network = multinet.multi_layer_network()

# Time-sliced layers
network.add_edges([
    ['Alice', '2020', 'Bob', '2020', 1],
    ['Alice', '2021', 'Bob', '2021', 1],
    ['Bob', '2021', 'Carol', '2021', 1],
    ['Alice', '2022', 'Carol', '2022', 1],
], input_type="list")
```

**Real-world applications:**

- Communication patterns over time
- Disease spread through populations
- Financial transaction networks
- Collaboration evolution

### Interdependent Networks

**Definition:** Multiple networks where the function of nodes in one layer depends on nodes in other layers.

**Characteristics:**

- **Dependency edges** encode functional relationships
- **Cascading failures** possible across layers
- **Critical infrastructure** applications

**Example:** Power grid (layer 1) depends on communication network (layer 2). If communication fails, power grid control fails.

**Real-world applications:**

- Infrastructure systems (power, water, communication)
- Supply chain networks
- Cyber-physical systems
- Ecological networks (species, habitats, resources)

## Inter-Layer Coupling

The **coupling strength**  $\omega$  controls how strongly layers are connected.

### Identity Coupling

The most common form connects each node to itself across layers:

$$E_C = \{((v, \alpha), (v, \beta)) : v \in V, \alpha \neq \beta \in L\}$$

with edge weight  $\omega$ .

**Choice of  $\omega$ :**

- $\omega = 1.0$  — Layers are equally important, nodes fully correspond
- $\omega < 1.0$  — Layers are loosely coupled, favor intra-layer paths
- $\omega > 1.0$  — Inter-layer transitions are encouraged

**Example:**

```
network = multinet.multi_layer_network()

# Add intra-layer edges
network.add_edges([
    ['A', 'layer1', 'B', 'layer1', 1],
    ['A', 'layer2', 'C', 'layer2', 1],
], input_type="list")

# Add inter-layer coupling (identity edges)
network.add_edges([
    ['A', 'layer1', 'A', 'layer2', 0.5], # omega = 0.5
], input_type="list")
```

## General Coupling

More complex couplings allow different nodes to connect across layers:

$$E_C = \{((v, \alpha), (w, \beta)) : (v, w) \in C_{\alpha\beta}\}$$

where  $C_{\alpha\beta}$  defines which nodes couple between layers  $\alpha$  and  $\beta$ .

## When to Use Multilayer Networks

Use multilayer modeling when:

### 1. Multiple relationship types have distinct semantics

Example: Friendship vs. professional collaboration—these networks have different properties and dynamics.

### 2. Node roles vary by context

Example: A person central in academic network may be peripheral in social media network.

### 3. Cross-layer interactions matter

Example: Information spreads from Twitter to traditional media—this cross-layer flow is meaningful.

### 4. Temporal evolution is important

Example: Community structure evolves over time—time-sliced layers preserve this evolution.

## 5. System-level resilience depends on layer dependencies

Example: Power grid failure affects communication, which affects emergency response.

## Choosing a Modeling Approach

### Decision Tree

1. Do relationships have fundamentally different types?  
YES → Use layers (multiplex or heterogeneous)  
NO → Use edge weights or attributes
2. Are the same entities present in all layers?  
YES → Multiplex network (strong coupling)  
NO → Heterogeneous network (weak/no coupling)
3. Does time evolution matter?  
YES → Temporal layers (time-sliced)  
NO → Static multilayer network
4. Are there functional dependencies between layers?  
YES → Interdependent network (dependency edges)  
NO → Standard multiplex/heterogeneous

### Common Mistakes

#### Over-aggregation

Combining layers that should stay separate (e.g., merging email and meeting networks loses information about mode transitions).

#### Under-aggregation

Creating too many sparse layers when edge weights would suffice (e.g., separate layer for each email vs. email timestamp as attribute).

#### Wrong coupling strength

Using  $\omega = 1.0$  when nodes don't fully correspond, or  $\omega \rightarrow 0$  when they do.

#### Mismatched identifiers

Using different IDs for the same entity across layers breaks multiplex structure.

### Why Flattening Fails

Flattening (aggregating all layers into a single graph) loses critical information:

#### 1. Community Structure

Multilayer communities may have:

- **Core in one layer, periphery in another**
- **Cross-layer bridges** that appear as spurious intra-layer connections when flattened

**Example:** Academic collaboration (dense group) + Twitter followers (different dense group). Flattening creates false bridges.

#### 2. Centrality

A node's importance depends on layer:

- **Structural centrality** in one layer (high degree)



- **Functional centrality** in another (betweenness for information flow)

Flattening mixes these distinct roles.

### 3. Path Structure

Meaningful paths often cross layers:

- Email → meeting → collaboration
- Twitter mention → news coverage → policy change

Flattening hides these cross-layer transitions.

### 4. Dynamics

Disease spreading, information diffusion, and cascading failures all depend on layer-specific parameters and inter-layer transitions. Flattening cannot capture:

- **Layer-specific transmission rates**
- **Mode-switching dynamics**
- **Asymmetric cross-layer effects**

## Key Terminology

### Intra-layer edges

Edges within a single layer, connecting  $(v, \alpha)$  to  $(w, \alpha)$ .

### Inter-layer edges

Edges between layers, connecting  $(v, \alpha)$  to  $(w, \beta)$  with  $\alpha \neq \beta$ .

### Node-layer pair

The tuple  $(v, \alpha)$  representing node  $v$  in layer  $\alpha$ —the fundamental unit of a multilayer network.

### Supra-adjacency matrix

The block matrix  $\mathbf{A}$  encoding all intra-layer and inter-layer edges.

### Coupling strength

The weight  $\omega$  of inter-layer edges, controlling how strongly layers interact.

### Multiplex network

Multilayer network with the same nodes in all layers.

### Heterogeneous information network

Multilayer network with different node types per layer.

## Working with Supra-Adjacency Matrices in Py3plex

```
from py3plex.core import multinet, random_generators

# Generate random multilayer network
network = random_generators.random_multilayer_ER(
    num_nodes=100,
    num_layers=3,
    probability=0.05,
    directed=False
)

# Get supra-adjacency matrix (sparse format)
supra_matrix = network.get_supra_adjacency_matrix()
```

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```
print(f"Matrix shape: {supra_matrix.shape}") # (300, 300) for 100 nodes x 3 layers
print(f"Matrix density: {supra_matrix.nnz / (supra_matrix.shape[0] ** 2):.4f}")

# Visualize matrix structure
network.visualize_matrix({"display": True})
```

The supra-adjacency matrix enables tensor-based algorithms and linear algebra operations on the full multilayer structure.

## Summary

Multilayer networks formalize systems with multiple relationship types by:

- Using **node-layer pairs** as the fundamental unit
- Encoding structure in the **supra-adjacency matrix**
- Supporting various types: **multiplex**, **heterogeneous**, **temporal**, **interdependent**
- Controlling interaction via **coupling strength**  $\omega$

Key takeaways:

1. **Multiplex** = same nodes, different edge types
2. **Heterogeneous** = different node types per layer
3. **Temporal** = time-sliced layers
4. **Coupling** = inter-layer connections (identity edges most common)
5. **Flattening loses information** — use multilayer analysis when layer structure matters

The next chapter explains how py3plex implements these concepts and why its design choices support efficient, correct multilayer analysis.

## Further Reading

- **Theory:** Kivela et al. (2014). “Multilayer networks.” *J. Complex Networks* 2(3): 203-271.
- **Physics:** Boccaletti et al. (2014). “The structure and dynamics of multilayer networks.” *Physics Reports* 544(1): 1-122.
- **Textbook:** Bianconi, G. (2018). *Multilayer Networks: Structure and Function*. Oxford University Press.
- **Applications:** De Domenico et al. (2013). “Mathematical formulation of multilayer networks.” *Physical Review X* 3(4): 041022.

## 1.7.3 Design of Py3plex

This chapter explains how py3plex is architected and why it makes specific design choices. Understanding these design principles will help you use the library more effectively and debug unexpected behavior.

### Architecture Overview

Py3plex is built on three foundational principles:

1. **NetworkX compatibility** — Leverage a mature, well-tested ecosystem
2. **Modular design** — Separate concerns for flexibility and extensibility

### 3. Simple, off-the-shelf functionality — Minimize barrier to entry

The library consists of several core modules:

```
py3plex/
├── core/                # Data structures (multi_layer_network class)
├── algorithms/          # Analysis methods
│   ├── statistics/      # Multilayer statistics
│   ├── centrality/      # Centrality measures
│   ├── community_detection/ # Community algorithms
│   └── paths/           # Path algorithms
├── dynamics/            # Dynamical processes (SIS, SIR, random walks)
├── dsl/                 # SQL-like query language
├── visualization/       # Plotting and visualization
├── io/                  # Data loading/saving
└── cli.py               # Command-line interface
```

## Node-Layer Pair Representation

### The Fundamental Unit

Py3plex represents multilayer networks using **node-layer pairs** as the fundamental unit. A node  $v$  in layer  $\alpha$  is stored as the tuple  $(v, \alpha)$ .

```
from py3plex.core import multinet

network = multinet.multi_layer_network()

# Add nodes explicitly as (node_id, layer_id) pairs
network.add_nodes([
    ('Alice', 'friends'),
    ('Bob', 'friends'),
    ('Alice', 'colleagues'),
])

# Or implicitly through edge addition
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Alice', 'colleagues', 'Bob', 'colleagues', 1],
], input_type="list")
```

### Why node-layer pairs?

- **Unambiguous identification** — The same logical entity (Alice) has distinct identities in different layers
- **NetworkX compatibility** — Tuples are valid NetworkX node identifiers
- **Efficient lookups** — Python's tuple hashing enables fast operations
- **Natural tensor mapping** — Node-layer pairs map directly to rows/columns of the supra-adjacency matrix

## The multi\_layer\_network Class

### Core Components

The `multi_layer_network` class wraps a NetworkX graph and adds multilayer-specific functionality:

```

network = multinet.multi_layer_network(directed=False)

# The underlying NetworkX graph (MultiGraph or MultiDiGraph)
G = network.core_network

# Layer management
layers = network.get_layers()           # ['friends', 'colleagues']
layer_map = network.layer_name_map      # {'friends': 0, 'colleagues': 1}

# Access node-layer pairs
nodes = list(network.get_nodes())       # [('Alice', 'friends'), ...]

```

#### Key attributes:

- `core_network` — The NetworkX graph storing all node-layer pairs and edges
- `layer_name_map` — Mapping from layer names to internal IDs
- `directed` — Whether the network is directed

### Multilayer vs. Multiplex Mode

Py3plex supports two operational modes:

#### Multilayer (default):

- General multilayer networks with arbitrary node sets per layer
- No automatic inter-layer edges
- Use for heterogeneous networks (different node types)

```

# Heterogeneous network: authors and papers
network = multinet.multi_layer_network(network_type='multilayer')
network.add_edges([
    ['Alice', 'authors', 'Paper1', 'papers', 1],
    ['Paper1', 'papers', 'ICML', 'venues', 1],
], input_type="list")

```

#### Multiplex:

- Same node set in all layers
- Automatically creates identity (coupling) edges with `type='coupling'`
- Use when the same entities appear in all layers

```

# Social network: same people, different relationships
network = multinet.multi_layer_network(network_type='multiplex')
network.load_network('social.edges', input_type='multiplex_edges')

# Coupling edges are auto-created: (Alice, friends) <-> (Alice, colleagues)

# Get explicit edges only (exclude coupling)
explicit_edges = list(network.get_edges(multiplex_edges=False))

```

## Relationship to NetworkX

### Direct Access to NetworkX

Py3plex does not hide NetworkX—it exposes the underlying graph for direct manipulation:

```
import networkx as nx

# Access the graph directly
G = network.core_network

# Use any NetworkX function
betweenness = nx.betweenness centrality(G)
communities = nx.community.louvain_communities(G)

# Modify the graph directly
G.nodes[('Alice', 'friends')][ 'age' ] = 30
```

#### Why this matters:

- You can use **any NetworkX algorithm** on the multilayer network
- You can **extend py3plex** by writing functions that operate on `core_network`
- You don't need to learn a completely new API—if you know NetworkX, you know most of py3plex

## NetworkX Interoperability

Convert between py3plex and NetworkX:

```
# From NetworkX to py3plex
import networkx as nx
G = nx.karate_club_graph()
network = multinet.multi_layer_network()
network.load_network_from_networkx(G, layer_name='social')

# From py3plex to NetworkX (extract a single layer)
friends_layer = network.get_layer_subgraph('friends')
```

This enables workflows that mix py3plex's multilayer capabilities with NetworkX's extensive algorithm library.

## Core Modules

### Algorithms

#### Statistics (`py3plex/algorithms/statistics/`)

Multilayer-specific metrics like degree distributions, layer overlap, and aggregation statistics.

#### Centrality (`py3plex/algorithms/centrality/`)

Centrality measures adapted for multilayer networks, including explainable centrality that breaks down scores by layer.

#### Community Detection (`py3plex/algorithms/community_detection/`)

Multilayer Louvain, Infomap, and modularity-based methods.

#### Paths (`py3plex/paths/`)

Shortest paths, random walks, and flow algorithms that respect layer structure.

### Dynamics

#### Dynamical Processes (py3plex/dynamics/)

OOP-style classes for epidemic models (SIS, SIR, SEIR), random walks, and custom dynamics. Each model has:

- `set_seed()` for reproducibility
- `run()` for execution
- `get_measure()` for extracting results

### DSL and Query Language

#### Domain-Specific Language (py3plex/dsl/)

SQL-like queries for network analysis:

```
from py3plex.dsl import Q, L

result = (
    Q.nodes()
    .from_layers(L["friends"])
    .where(degree__gt=5)
    .compute("betweenness centrality")
    .execute(network)
)
```

The DSL provides:

- **Builder API** — Chainable, type-hinted query construction
- **String syntax** — SQL-like queries as strings
- **EXPLAIN mode** — Query execution plans
- **Export** — Results to pandas, JSON, CSV

### Visualization

#### Visualization (py3plex/visualization/)

Publication-ready plots including:

- **Hairball plots** — 2D/3D force-directed layouts with layer separation
- **Matrix plots** — Supra-adjacency matrix visualization
- **Diagonal layouts** — Specialized for large multilayer networks

### I/O

#### Input/Output (py3plex/io/)

Support for multiple formats:

- **Edgelists** — Simple text format
- **GraphML** — XML-based format
- **Arrow/Parquet** — High-performance columnar format for large networks
- **JSON** — Flexible structured format

## Design Principles in Practice

### 1. Flexible Input

Multiple ways to accomplish the same task:

```
# Method 1: List format
network.add_edges([[ 'A', 'L1', 'B', 'L1', 1]], input_type="list")

# Method 2: Dict format (explicit)
network.add_edges([{'source': 'A', 'target': 'B',
                    'source_type': 'L1', 'target_type': 'L1'}])

# Method 3: Load from file
network.load_network("data.edgelist", input_type="edgelist")
```

### 2. Lazy Evaluation

Expensive operations are computed only when needed:

```
# Network creation and edge addition are fast
network = multinet.multi_layer_network()
network.add_edges([...]) # No supra-adjacency matrix computed

# Matrix is computed only when requested
supra_adj = network.get_supra_adjacency_matrix() # Computed here
```

### 3. Graceful Degradation

The library handles edge cases and provides informative warnings:

- Missing layers → created automatically
- Empty networks → return sensible defaults
- Invalid parameters → clear error messages

### 4. Extensibility

Adding new algorithms is straightforward:

```
def my_custom_metric(network):
    """Custom multilayer metric."""
    G = network.core_network
    # Implement your algorithm using G
    return result

# Use immediately
score = my_custom_metric(network)
```

### Why These Choices Matter

**NetworkX compatibility** means you can:

- Use hundreds of existing algorithms without modification
- Mix py3plex with other NetworkX-based tools
- Leverage a mature, well-documented ecosystem

**Node-layer pair representation** ensures:

- Unambiguous node identity across layers
- Efficient graph operations (hashing, lookups)
- Direct mapping to mathematical definitions

**Modular architecture** allows:

- Importing only needed functionality
- Easy testing and debugging
- Clean separation of concerns

**Multiple input methods** accommodate:

- Different data sources and formats
- User preferences and workflows
- Programmatic and interactive use

### Limitations and Trade-offs

No design is perfect. Py3plex makes conscious trade-offs:

#### Memory:

The supra-adjacency matrix for large networks can be memory-intensive. Use sparse matrix formats (default) and avoid materializing the full matrix when possible.

#### Performance:

Some operations (e.g., cross-layer random walks) require iteration over node-layer pairs, which is slower than pure NetworkX on single-layer graphs.

#### Coupling semantics:

The automatic coupling in multiplex mode assumes identity edges. For more complex inter-layer relationships, use multilayer mode and add edges explicitly.

### Summary

Py3plex is designed around:

1. **Node-layer pairs** as the fundamental unit
2. **NetworkX compatibility** for interoperability and extensibility
3. **Modular architecture** for flexibility
4. **Simple, practical APIs** for ease of use

These principles enable:

- Correct multilayer analysis (proper layer semantics)
- Efficient implementation (leverage NetworkX)



- Extensible design (add your own algorithms)
- Practical workflows (multiple input formats, visualization, DSL)

The following chapters will show how to use these design elements in practice: loading data, visualizing networks, and running multilayer-specific algorithms.

### Further Reading

- NetworkX documentation: <https://networkx.org/documentation/stable/>
- Py3plex API reference: *Appendix E: Extended API and DSL Reference*
- Code architecture: *Appendix A: Repository Layout and Scripts*

## 1.7.4 Installation and Getting Started

This chapter covers installing py3plex and running your first multilayer network analysis. We provide a minimal quick-start example followed by installation details for different use cases.

### DSL Quick Start

After installation, try the DSL for instant network exploration:

```
from py3plex.core import multinet
from py3plex.dsl import Q, L

# Create a simple network
network = multinet.multi_layer_network()
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Bob', 'friends', 'Carol', 'friends', 1],
], input_type="list")

# Query it with DSL
result = (
    Q.nodes()
    .compute("degree")
    .order_by("-degree")
    .execute(network)
)
print(result.to_pandas())
```

DSL makes network analysis intuitive from day one!

### Quick Install

For most users, installation is straightforward:

```
pip install py3plex
```

This installs py3plex with core dependencies (NetworkX, NumPy, SciPy, pandas, matplotlib).

**Verify installation:**

```
import py3plex
print(py3plex.__version__) # Should print 1.x
```

### System Requirements

**Python:** 3.8 or higher (tested on 3.8, 3.9, 3.10, 3.11, 3.12)

**Platforms:** Linux, macOS, Windows

**Core dependencies** (installed automatically):

- NetworkX (2.5) — Graph data structures
- NumPy (0.8) — Numerical computing
- SciPy (1.1.0) — Sparse matrices
- pandas — Data manipulation
- matplotlib — Visualization
- scikit-learn — ML utilities

### Hello Multilayer Network

Create and analyze a multilayer network in 30 seconds:

```
from py3plex.core import multinet

# Create network
network = multinet.multi_layer_network()

# Add edges: [source, source_layer, target, target_layer, weight]
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Bob', 'friends', 'Carol', 'friends', 1],
    ['Alice', 'colleagues', 'Bob', 'colleagues', 1],
    ['Bob', 'colleagues', 'Dave', 'colleagues', 1],
], input_type="list")

# Display statistics
network.basic_stats()
```

#### Output:

```
Number of nodes: 6
Number of edges: 4
Number of unique nodes (as node-layer tuples): 6
Number of unique node IDs (across all layers): 4
Nodes per layer:
  Layer 'friends': 3 nodes
  Layer 'colleagues': 3 nodes
```

**Key insight:** The network has 6 node-layer pairs (Alice-friends, Alice-colleagues, Bob-friends, Bob-colleagues, Carol-friends, Dave-colleagues) but only 4 unique people.

### Visualize the Network

```
from py3plex.visualization.multilayer import draw_multilayer_default
```

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```
# Simple visualization
draw_multilayer_default([network], display=True)
```

This creates a force-directed layout with nodes colored by layer.

## Basic Analysis

```
from py3plex.algorithms.statistics import multilayer_statistics as mls

# Layer density (how connected is each layer?)
print(f"Friends density: {mls.layer_density(network, 'friends'):.3f}")
# Output: Friends density: 0.667 (2 of 3 possible edges exist)

# Node activity (fraction of layers a node appears in)
print(f"Bob's activity: {mls.node_activity(network, 'Bob'):.3f}")
# Output: Bob's activity: 1.000 (appears in both layers)
```

## Query with DSL

Use SQL-like queries to explore the network:

```
from py3plex.dsl import execute_query

# Find high-degree nodes
result = execute_query(network, 'SELECT nodes WHERE degree > 1')
print(f"Found {result['count']} nodes with degree > 1")

# Get all nodes in a specific layer
result = execute_query(network, 'SELECT nodes WHERE layer="friends"')
```

The DSL (covered in Part III) provides a powerful way to filter, compute, and analyze multilayer networks without writing explicit loops.

## Installation Options

### Development Installation

For contributors or to access the latest features:

```
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex
pip install -e .
```

The `-e` flag installs in editable mode—changes to source code are immediately available.

### For development with testing/linting tools:

```
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex
make setup          # Create virtual environment
make dev-install    # Install with dev dependencies
```

This installs pytest, black, ruff, mypy, and sphinx for development workflows.

### Docker Installation

For a containerized environment with all dependencies:

```
# Clone repository
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex

# Build image
docker build -t py3plex:latest .

# Run commands
docker run --rm py3plex:latest --version
```

#### Benefits:

- No Python setup required
- Consistent environment across systems
- Isolated from system Python

### Optional Dependencies

Py3plex supports optional feature sets:

```
# Advanced community detection (Infomap)
pip install py3plex[infomap]

# Extra algorithms (Louvain, cdlib)
pip install py3plex[algos]

# Advanced visualization (Plotly, igraph)
pip install py3plex[viz]

# High-performance I/O (Arrow/Parquet)
pip install py3plex[arrow]

# Install multiple extras
pip install py3plex[infomap,viz,algos,arrow]
```

### Virtual Environments

We strongly recommend using virtual environments:

#### Using venv:

```
python3 -m venv py3plex-env
source py3plex-env/bin/activate # Linux/macOS
# py3plex-env\Scripts\activate # Windows
pip install py3plex
```

#### Using conda:

```
conda create -n py3plex python=3.10
conda activate py3plex
pip install py3plex
```

## Key Concepts

Before proceeding, understand these fundamental concepts:

### Node-Layer Pairs

In py3plex, a node is always associated with a layer. The tuple (node\_id, layer\_id) is the fundamental unit.

- ('Alice', 'friends') is different from ('Alice', 'colleagues')
- This allows the same logical entity to have different properties and connections in different contexts

Example:

```
# Access nodes as (node_id, layer_id) tuples
for node in network.get_nodes():
    print(node) # ('Alice', 'friends'), ('Bob', 'friends'), ...
```

### NetworkX Compatibility

Py3plex uses NetworkX as its underlying graph representation:

```
import networkx as nx

# Direct access to the NetworkX graph
G = network.core_network

# Use any NetworkX algorithm
betweenness = nx.betweenness centrality(G)

# Modify directly
G.nodes[('Alice', 'friends')]['age'] = 30
```

This means:

- Any NetworkX algorithm works on py3plex networks
- You can mix py3plex and NetworkX functions freely
- Learning curve is minimal if you already know NetworkX

### Input Formats

Py3plex supports multiple edge input formats:

**List format** (used in examples above):

```
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
], input_type="list")
```

**Dictionary format** (explicit, Pythonic):

```
network.add_edges([[
    'source': 'Alice',
    'source_type': 'friends',
    'target': 'Bob',
    'target_type': 'friends',
    'weight': 1
]])
```

**File loading** (for larger networks):

```
# Edgelist format
network.load_network("data.edgelist", input_type="edgelist")

# GraphML, JSON, Arrow, etc.
# See Chapter 5 for details
```

### Common Issues

#### Issue: “No module named ‘py3plex’”

**Solution:** Ensure py3plex is installed in the active Python environment:

```
pip install py3plex
```

Verify which Python interpreter you’re using:

```
which python # Linux/macOS
where python # Windows
```

#### Issue: Import errors for optional dependencies

**Solution:** Install the relevant extra:

```
pip install py3plex[infomap] # For Infomap
pip install py3plex[viz]     # For Plotly
```

#### Issue: Permission denied on Linux/macOS

**Solution:** Use a virtual environment (recommended) or install with `--user`:

```
pip install --user py3plex
```

### Verifying Installation

Run a self-test to confirm everything works:

```
from py3plex.core import multinet
from py3plex.algorithms.statistics import multilayer_statistics as mls

# Create test network
network = multinet.multi_layer_network()
network.add_edges([
    ['A', 'L1', 'B', 'L1', 1],
```

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```

    ['B', 'L1', 'C', 'L1', 1],
], input_type="list")

# Test basic operations
assert network.get_number_of_nodes() == 3
assert mls.layer_density(network, 'L1') > 0

print("✓ py3plex is working correctly")

```

## Next Steps

Now that you have py3plex installed and understand the basics:

- **Chapter 5** — Learn about data loading, supported formats, and best practices for representing multilayer networks
- **Chapter 6** — Explore visualization techniques for multilayer networks
- **Chapter 7** — Apply core algorithms: community detection, centrality measures, and dynamics

### For immediate exploration:

Browse the `examples/` directory in the repository for 50+ working examples covering various use cases.

## Summary

This chapter covered:

1. **Quick install** — `pip install py3plex`
2. **Hello World** — Create a multilayer network in 30 seconds
3. **Basic analysis** — Statistics, visualization, and DSL queries
4. **Installation options** — Development, Docker, optional dependencies
5. **Key concepts** — Node-layer pairs, NetworkX compatibility, input formats

With py3plex installed, you're ready to analyze multilayer networks. The next chapter covers loading real data and representing complex multilayer structures.

## 1.7.5 Data Loading and Representation

This chapter covers how to load multilayer networks from various data sources, choose appropriate formats, and represent complex network structures correctly.

### DSL Tip: Validate Data After Loading

Use DSL to quickly validate loaded data:

```

from py3plex.dsl import Q, L

# Check layer sizes
for layer in network.get_layers():
    count = Q.nodes().from_layers(L[layer]).execute(network).count
    print(f"{layer}: {count} nodes")

# Find potential data issues

```

```
isolated = Q.nodes().where(degree=0).execute(network)
if isolated.count > 0:
    print(f"Warning: {isolated.count} isolated nodes")

# Verify high-degree nodes make sense
hubs = (
    Q.nodes()
    .where(degree__gt=20)
    .compute("degree")
    .execute(network)
)
print(f"Hubs (degree > 20): {hubs.count}")
```

Quick validation catches data issues early!

## Data Loading Basics

### Multiple Input Methods

Py3plex supports several ways to create multilayer networks:

#### 1. Direct edge addition (for programmatic construction):

```
from py3plex.core import multinet

network = multinet.multi_layer_network()
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Bob', 'friends', 'Carol', 'friends', 1],
], input_type="list")
```

#### 2. File loading (for external data):

```
# Load from file
network = multinet.multi_layer_network()
network.load_network("data.edgelist", input_type="edgelist")
```

#### 3. From NetworkX (for integration):

```
import networkx as nx

G = nx.karate_club_graph()
network = multinet.multi_layer_network()
network.load_network_from_networkx(G, layer_name='social')
```

## Edgelist Format

The **edgelist** format is the simplest and most common:

```
# File: network.edgelist
Alice friends Bob friends 1.0
Bob friends Carol friends 1.0
Alice colleagues Bob colleagues 0.8
```



**Format:** source source\_layer target target\_layer weight

**Loading:**

```
network = multinet.multi_layer_network()
network.load_network("network.edgelist", input_type="edgelist")
```

**Best practices:**

- Use consistent node identifiers across layers
- Separate fields with spaces or tabs
- Include weights (default to 1.0 if omitted)
- Comment lines start with #

## Dictionary Format

For programmatic construction, the dictionary format is most explicit:

```
network.add_edges([
    {
        'source': 'Alice',
        'source_type': 'friends',
        'target': 'Bob',
        'target_type': 'friends',
        'weight': 1.0,
        'timestamp': '2023-01-15' # Optional attributes
    },
    {
        'source': 'Bob',
        'source_type': 'colleagues',
        'target': 'Carol',
        'target_type': 'colleagues',
        'weight': 0.8
    }
])
```

**Advantages:**

- Self-documenting (field names explicit)
- Supports arbitrary edge attributes
- Type-safe (Python dicts)
- Pythonic and readable

## Modern I/O System

Py3plex provides a modern I/O system (`py3plex.io`) for high-performance serialization:

## Supported Formats

| Format  | Extension | Best For                          | Trade-offs           |
|---------|-----------|-----------------------------------|----------------------|
| JSON    | .json     | Human-readable, small networks    | Slower, larger files |
| JSONL   | .jsonl    | Streaming, large networks         | Not human-friendly   |
| CSV     | .csv      | Spreadsheet tools, manual editing | Limited attributes   |
| Arrow   | .arrow    | High performance, large networks  | Requires pyarrow     |
| Parquet | .parquet  | Storage, compression, archiving   | Requires pyarrow     |

## Reading and Writing

```
from py3plex.io import read, write, MultiLayerGraph

# Write (auto-detects format from extension)
write(network, 'network.json')
write(network, 'network.arrow')
write(network, 'network.parquet')

# Read (auto-detects format)
graph = read('network.json')
graph = read('network.arrow')

# Specify format explicitly
write(network, 'myfile.dat', format='json')
graph = read('myfile.dat', format='parquet')
```

## Apache Arrow Format (Recommended for Large Networks)

Arrow provides 2-3x faster I/O and better compression:

```
# Install Arrow support
pip install 'py3plex[arrow]'
```

```
from py3plex.io import read, write

# Feather format (fast, uncompressed)
write(graph, 'network.arrow')
graph = read('network.arrow')

# Parquet format (compressed, archival)
write(graph, 'network.parquet')
graph = read('network.parquet')
```

**Performance comparison** (1000 nodes, 5000 edges):

| Format  | Write Time | Read Time | File Size |
|---------|------------|-----------|-----------|
| Arrow   | 0.016s     | 0.008s    | 0.46 MB   |
| Parquet | 0.020s     | 0.010s    | 0.35 MB   |
| JSON    | 0.046s     | 0.030s    | 1.09 MB   |

#### When to use Arrow:

- Large networks (>10k nodes)
- Performance-critical pipelines
- Interoperability with pandas, Spark, DuckDB
- Production data workflows

#### When to use JSON:

- Small networks (<1k nodes)
- Human readability required
- Debugging and manual editing
- Maximum compatibility

### CSV with Sidecars

CSV format supports optional sidecar files for attributes:

```
# Write with sidecars
write(graph, 'edges.csv', write_sidecars=True)
# Creates: edges.csv, nodes.csv, layers.csv

# Read with sidecars
graph = read('edges.csv',
             nodes_file='nodes.csv',
             layers_file='layers.csv')
```

### Data Validation and Best Practices

#### Always Verify After Loading

After loading external data, always verify structure:

```
network.load_network("data.edgelist", input_type="edgelist")

# Verify structure
print(f"Layers: {network.get_layers()}")
print(f"Nodes per layer: {network.get_number_of_nodes_per_layer()}")

# Check for unexpected layers
expected_layers = {'friends', 'colleagues'}
actual_layers = set(network.get_layers())
if actual_layers != expected_layers:
    print(f"Warning: unexpected layers {actual_layers - expected_layers}")
```

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```
# Basic statistics
network.basic_stats()
```

### Consistent Node Identifiers

**Rule:** The same entity must have the same ID across all layers.

**Good:**

```
# Alice has the same ID in both layers
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Alice', 'colleagues', 'Carol', 'colleagues', 1],
], input_type="list")
```

**Bad:**

```
# Alice's ID differs across layers-breaks multiplex structure
network.add_edges([
    ['alice_social', 'friends', 'Bob', 'friends', 1],
    ['Alice_Work', 'colleagues', 'Carol', 'colleagues', 1],
], input_type="list")
```

### Naming Conventions

**Layer names:**

- Use descriptive names: 'coauthor', 'twitter', not 'layer1', 'l2'
- Lowercase with underscores: 'social\_media', not 'SocialMedia'
- Avoid special characters

**Node IDs:**

- Use stable, meaningful identifiers (user IDs, DOIs, names)
- Avoid auto-generated IDs that change across runs
- Preserve external IDs when loading from databases

### Metadata and Attributes

Store metadata as node/edge attributes:

```
# Add node attributes
network.core_network.nodes[('Alice', 'friends')]['age'] = 30
network.core_network.nodes[('Alice', 'friends')]['country'] = 'USA'

# Add edge attributes
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1, {'type': 'close', 'since': 2018}]
], input_type="list")
```

**Recommended practices:**

- Use consistent attribute names across layers

- Store timestamps as ISO format strings: '2023-01-15T10:30:00Z'
- Use numeric types for attributes that will be analyzed
- Document attribute semantics in code comments

## Representing Different Network Types

### Multiplex Networks (Same Nodes Across Layers)

For networks where the same entities appear in all layers:

```
network = multinet.multi_layer_network(network_type='multiplex')

# Load data
network.load_network('social.edges', input_type='multiplex_edges')

# Coupling edges are automatically created
# (Alice, friends) <-> (Alice, colleagues)

# Get explicit edges only (exclude coupling)
explicit_edges = list(network.get_edges(multiplex_edges=False))
```

When to use:

- Same people across multiple social platforms
- Same cities in transportation modes (air, rail, road)
- Same proteins in different tissue types

### Heterogeneous Networks (Different Node Types)

For networks with different entity types per layer:

```
network = multinet.multi_layer_network(network_type='multilayer')

# Different node types
network.add_edges([
    ['Alice', 'authors', 'Paper1', 'papers', 1],
    ['Bob', 'authors', 'Paper1', 'papers', 1],
    ['Paper1', 'papers', 'ICML', 'venues', 1],
], input_type="list")
```

When to use:

- Academic networks (authors, papers, venues)
- E-commerce (users, products, sellers)
- Biomedical (drugs, diseases, targets)

### Temporal Networks (Time-Sliced Layers)

For networks that evolve over time:

```
network = multinet.multi_layer_network()
```

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```
# Time windows as layers
network.add_edges([
    ['Alice', '2020', 'Bob', '2020', 1],
    ['Alice', '2021', 'Bob', '2021', 1],
    ['Bob', '2021', 'Carol', '2021', 1],
], input_type="list")
```

#### Best practices:

- Use consistent time granularity (years, months, days)
- Layer names should be sortable: '2020-01', '2020-02', ...
- Add temporal edges connecting adjacent time slices if modeling evolution

## Converting Between Formats

### NetworkX to Py3plex

```
import networkx as nx
from py3plex.core import multinet

# Create NetworkX graph
G = nx.karate_club_graph()

# Convert to py3plex
network = multinet.multi_layer_network()
network.load_network_from_networkx(G, layer_name='social')
```

### Py3plex to NetworkX

```
# Extract single layer as NetworkX graph
friends_layer = network.get_layer_subgraph('friends')

# Or access the full graph directly
full_graph = network.core_network # MultiGraph or MultiDiGraph
```

### Pandas DataFrame to Py3plex

```
import pandas as pd

# Load edges from DataFrame
df = pd.read_csv('edges.csv')
# Columns: source, source_layer, target, target_layer, weight

# Convert to list of lists
edges = df[['source', 'source_layer', 'target', 'target_layer', 'weight']].values.
    ↪ tolist()

network = multinet.multi_layer_network()
network.add_edges(edges, input_type="list")
```

## Common Data Loading Patterns

### Pattern 1: Load from Multiple Files

```
network = multinet.multi_layer_network()

# Load different layers from different files
for layer_name, filename in [('friends', 'friends.txt'),
                             ('colleagues', 'colleagues.txt')]:
    with open(filename) as f:
        for line in f:
            source, target, weight = line.strip().split()
            network.add_edges([
                [source, layer_name, target, layer_name, float(weight)]
            ], input_type="list")
```

### Pattern 2: Load from Database

```
import sqlite3

conn = sqlite3.connect('network.db')
cursor = conn.execute(
    "SELECT source, source_layer, target, target_layer, weight FROM edges"
)

network = multinet.multi_layer_network()
edges = [[row[0], row[1], row[2], row[3], row[4]] for row in cursor]
network.add_edges(edges, input_type="list")
```

### Pattern 3: Incremental Loading (Large Networks)

```
network = multinet.multi_layer_network()

# Load in batches to manage memory
BATCH_SIZE = 10000

with open('large_network.edgelist') as f:
    batch = []
    for line in f:
        parts = line.strip().split()
        batch.append(parts)

        if len(batch) >= BATCH_SIZE:
            network.add_edges(batch, input_type="list")
            batch = []

# Add remaining edges
if batch:
    network.add_edges(batch, input_type="list")
```

## Troubleshooting

### Issue: Duplicate Edges

**Symptom:** More edges than expected after loading.

**Solution:** Check for duplicate entries in source data:

```
# Count edge multiplicity
edge_counts = {}
for edge in network.get_edges():
    edge_counts[edge] = edge_counts.get(edge, 0) + 1

duplicates = {e: c for e, c in edge_counts.items() if c > 1}
if duplicates:
    print(f"Found {len(duplicates)} duplicate edges")
```

### Issue: Unexpected Layers

**Symptom:** Layers you didn't expect appear in the network.

**Solution:** Verify layer names in source data:

```
expected = {'friends', 'colleagues'}
actual = set(network.get_layers())

if actual != expected:
    print(f"Unexpected layers: {actual - expected}")
    print(f"Missing layers: {expected - actual}")
```

### Issue: Node ID Mismatches

**Symptom:** Same entity appears as different nodes across layers.

**Solution:** Normalize node IDs before loading:

```
def normalize_id(node_id):
    """Normalize node identifiers."""
    return str(node_id).strip().lower()

# Apply during loading
edges = [[normalize_id(s), sl, normalize_id(t), tl, w]
         for s, sl, t, tl, w in raw_edges]
network.add_edges(edges, input_type="list")
```

## Summary

This chapter covered:

1. **Loading methods** — Direct addition, files, NetworkX import
2. **Data formats** — Edgelist, dictionary, JSON, CSV, Arrow/Parquet
3. **Best practices** — Consistent IDs, validation, metadata storage
4. **Network types** — Multiplex, heterogeneous, temporal
5. **Common patterns** — Multi-file loading, databases, incremental loading



**Key recommendations:**

- Use **Arrow/Parquet** for large networks (>10k nodes)
- Use **JSON** for small networks and debugging
- **Always verify** structure after loading
- Use **consistent node IDs** across layers
- Store **metadata as attributes** for richer analysis

The next chapter covers visualization techniques for exploring multilayer network structure.

**Further Reading**

- Arrow format specification: <https://arrow.apache.org/docs/>
- NetworkX I/O reference: <https://networkx.org/documentation/stable/reference/readwrite/>
- Chapter 7 (Core Algorithms) for analysis after loading

**1.7.6 Visualization and Exploration**

*TODO: This chapter will be expanded from docfiles/visualization\_guide.rst and user\_guide/visualization.rst*

**Overview**

[Introduce multilayer network visualization challenges and py3plex's capabilities]

**Core Visualization Capabilities****Static Visualizations**

[Hairball plots, force-directed layouts, layer separation]

```
from py3plex.visualization.multilayer import draw_multilayer_default

# Basic visualization
draw_multilayer_default([network], display=True)
```

**Interactive Visualizations**

[Plotly integration, 3D layouts]

**Matrix Visualizations**

[Supra-adjacency matrix visualization]

```
network.visualize_matrix({"display": True})
```

**Visualization Best Practices****When to Use Each Layout**

[Guidelines for choosing layouts based on network size and structure]

### Color Schemes and Aesthetics

[Layer coloring, node attributes, edge styling]

### Performance Considerations

[Size limits, performance tips, when visualization breaks down]

### Exploration Workflows

#### Layer Views

[Examining individual layers]

#### Cross-Layer Patterns

[Visualizing inter-layer connections]

#### Node Neighborhoods

[Ego networks, subgraph extraction]

### Summary

[Key points about multilayer visualization]

**Next chapter:** Core algorithms for multilayer analysis

*Source files to integrate:* - docfiles/visualization\_guide.rst - docfiles/user\_guide/visualization.rst - docfiles/r Ricci\_visualization.rst - examples/visualization/\*

## 1.7.7 Core Algorithms: Communities, Centrality, Dynamics

This chapter covers py3plex's three major algorithm families for multilayer network analysis: community detection, centrality measures, and dynamics/spreading processes.

### DSL Tip: Streamline Algorithm Workflows

The DSL simplifies many algorithmic workflows:

```
from py3plex.dsl import Q, L

# Quick centrality analysis
top_central = (
    Q.nodes()
    .from_layers(L["social"])
    .compute("betweenness centrality", "pagerank", "degree")
    .order_by("-betweenness centrality")
    .limit(20)
    .execute(network)
)

# Multi-layer comparison
for layer in ["social", "work"]:
    result = (
        Q.nodes()
```

```

        .from_layers(L[layer])
        .compute("betweenness centrality")
        .execute(network)
    )
    print(f"{layer}: avg BC = {sum(result.measures['betweenness centrality'].
→values()) / result.count:.4f}")

```

See Chapter 8 for comprehensive DSL coverage!

## Overview

[Introduce three major algorithm families for multilayer analysis]

## Community Detection

### Multilayer Modularity

[Mathematical definition and implementation]

```

from py3plex.algorithms.community_detection import multilayer_louvain

# Detect communities
communities = multilayer_louvain.best_partition(network.core_network)

```

## Algorithms Available

[Louvain, Infomap, Label Propagation]

**Complexity:**  $O(n \log n)$  for most algorithms

## Choosing a Community Detection Method

[Guidelines based on network size, structure, and goals]

## Centrality Measures

### Multilayer PageRank

[Random walk-based centrality]

### Degree Centrality

[Intra-layer vs inter-layer degree]

### Betweenness Centrality

[Path-based importance]

## Explainable Centrality

[Breaking down centrality scores by layer]

```
from py3plex.algorithms.centralities.explain import explain_node centrality

# Get layer-wise breakdown
explanation = explain_node centrality(network, 'Alice', measure='degree')
```

## Dynamics and Processes

py3plex provides comprehensive support for simulating dynamical processes on multilayer networks, including epidemic models and random walks.

**For complete documentation**, see the user guide section on multilayer dynamics and the SIR epidemic simulator documentation.

## Random Walks

Random walks simulate exploration and diffusion on networks:

```
from py3plex.dynamics import RandomWalkDynamics

# Create random walk
walk = RandomWalkDynamics(network, start_node=0, lazy_probability=0.1)
walk.set_seed(42)
results = walk.run(steps=1000)

# Analyze trajectory
trajectory = results.get_measure("trajectory")
```

Random walks are fundamental to PageRank, community detection, and network embeddings (Node2Vec, DeepWalk).

## Epidemic Models

Simulate disease spread and information diffusion with SIR and SIS models:

```
from py3plex.dynamics import SIRDynamics, SISDynamics

# SIR: Susceptible-Infected-Recovered (with immunity)
sir = SIRDynamics(network, beta=0.3, gamma=0.1, initial_infected=0.1)
sir.set_seed(42)
results = sir.run(steps=100)

# Extract measures
prevalence = results.get_measure("prevalence")
state_counts = results.get_measure("state_counts")

print(f"Peak prevalence: {prevalence.max():.2%}")
print(f"Final recovered: {state_counts['R'][-1]:.2%}")
```

### Key parameters:

- **beta ()**: Transmission rate (infection probability per contact)
- **gamma ()**: Recovery rate (recovery probability per time step)
- **$R_0 = \beta / \gamma$** : Basic reproduction number (epidemic threshold at  $R_0=1$ )

### Models:

- **SIR** — Epidemic with immunity (measles, chickenpox). Always dies out eventually.
- **SIS** — Endemic disease without immunity (common cold, malware). Reaches stable equilibrium.

## Multilayer Dynamics

Infection spreads through multiple interaction channels:

```
# Two-layer network: physical + digital contacts
network = multinet.multi_layer_network(directed=False)

# Add physical layer (local connections)
# Add digital layer (global connections)

# Run dynamics
sir = SIRdynamics(network, beta=0.3, gamma=0.1)
results = sir.run(steps=100)
```

The multilayer structure allows simultaneous spread through different contexts, often leading to faster and more extensive diffusion than single-layer networks.

## Algorithm Complexity and Scaling

[Performance characteristics, memory requirements, when to use each]

## Summary

[Recap of algorithm families and when to use each]

**Next chapter:** Introduction to the py3plex DSL

Source files to integrate: - docfiles/user\_guide/community\_detection.rst - docfiles/user\_guide/statistics.rst  
 - docfiles/tutorials/multilayer\_centrality.rst - docfiles/tutorials/explainable\_centrality.rst  
 - docfiles/sir\_epidemic\_simulator.rst - DYNAMICS\_IMPLEMENTATION.md - docfiles/multilayer\_centrality\_matrix\_functions.rst - docfiles/ricci\_curvature.rst

## 1.7.8 Introduction to the Py3plex DSL

This chapter introduces the py3plex Domain-Specific Language (DSL), a SQL-like query language for expressing multilayer network analyses concisely. The DSL is a **major first-class feature** that sets py3plex apart from other network libraries.

### DSL at a Glance

The DSL provides intuitive SQL-like syntax for network queries:

```
from py3plex.dsl import Q, L
```

```
# Simple: Find high-degree nodes
```

```
result = Q.nodes().where(degree__gt=5).execute(network)
```

```
# Powerful: Multi-layer analysis with export
```

```
result = (
    Q.nodes()
    .from_layers(L["social"] + L["work"])
    .where(degree__gt=3)
```

```
.compute("betweenness centrality", "pagerank")
.order_by("-betweenness centrality")
.limit(20)
.export_csv("top_influencers.csv")
.execute(network)
)
```

Express complex analyses in a few readable lines!

## Why a DSL for Networks?

Traditional network analysis requires writing explicit loops and conditionals:

```
# Traditional approach: verbose and error-prone
high_degree_nodes = []
for node in network.get_nodes():
    if node[1] == 'social': # Check layer
        degree = network.core_network.degree(node)
        if degree > 5:
            high_degree_nodes.append(node)

# Compute centrality for filtered nodes
centralities = {}
for node in high_degree_nodes:
    centralities[node] = nx.betweenness centrality(G)[node]
```

The DSL provides a declarative alternative:

```
# DSL approach: clear and concise
from py3plex.dsl import execute_query

result = execute_query(network,
    'SELECT nodes WHERE layer="social" AND degree > 5 '
    'COMPUTE betweenness centrality'
)
```

### Benefits:

1. **Declarative** — State *what* you want, not *how* to get it
2. **Readable** — SQL-like syntax is familiar and self-documenting
3. **Composable** — Build complex queries from simple building blocks
4. **Maintainable** — Less code means fewer bugs

## DSL Versions

Py3plex provides two DSL interfaces:

### String DSL (v1)

SQL-like queries as strings. Good for simple queries and REPL exploration.

```
result = execute_query(network, 'SELECT nodes WHERE degree > 5')
```

### Builder API (v2, recommended)

Python-native chainable API with type hints and autocompletion.

```
from py3plex.dsl import Q, L

result = (
    Q.nodes()
    .from_layers(L["social"])
    .where(degree__gt=5)
    .execute(network)
)
```

This chapter focuses on the **Builder API (v2)** as it provides a better development experience. The string DSL is covered briefly for reference.

## Basic Query Structure

A typical DSL query has four parts:

1. **Target** — What to select (nodes or edges)
2. **Layer filtering** — Which layers to include (optional)
3. **Conditions** — Filter criteria (optional)
4. **Measures** — What to compute (optional)

Example:

```
from py3plex.dsl import Q, L

result = (
    Q.nodes()                                # 1. Select nodes
    .from_layers(L["social"])                 # 2. Filter to social layer
    .where(degree__gt=5)                     # 3. Filter by degree > 5
    .compute("betweenness centrality")       # 4. Compute centrality
    .execute(network)                       # Execute query
)
```

## Your First Query

Let's build a simple query step-by-step.

**Setup:**

```
from py3plex.core import multinet
from py3plex.dsl import Q, L

# Create network
network = multinet.multi_layer_network()
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Bob', 'friends', 'Carol', 'friends', 1],
    ['Carol', 'friends', 'Dave', 'friends', 1],
    ['Alice', 'colleagues', 'Bob', 'colleagues', 1],
    ['Bob', 'colleagues', 'Eve', 'colleagues', 1],
], input_type="list")
```

**Query 1: Select all nodes**

```
result = Q.nodes().execute(network)

print(f"Count: {result.count}")
# Output: Count: 7

for node in result:
    print(node)
# Output: ('Alice', 'friends'), ('Bob', 'friends'), ...
```

#### Query 2: Filter by layer

```
result = Q.nodes().from_layers(L["friends"]).execute(network)

print(f"Nodes in friends layer: {result.count}")
# Output: Nodes in friends layer: 4
```

#### Query 3: Filter by degree

```
result = Q.nodes().where(degree__gt=1).execute(network)

print(f"High-degree nodes: {result.count}")
# Output: High-degree nodes: 2 (Bob in both layers)
```

#### Query 4: Compute centrality

```
result = (
    Q.nodes()
    .from_layers(L["friends"])
    .compute("betweenness centrality")
    .execute(network)
)

for node, centrality in result.measures['betweenness centrality'].items():
    print(f"{node}: {centrality:.3f}")
# Output: ('Bob', 'friends'): 1.000, others: 0.000
```

### Layer Filtering

The DSL provides powerful layer algebra operations.

#### DSL Layer Algebra

Layer algebra lets you combine layers with set operations:

```
from py3plex.dsl import Q, L

# Union: nodes in social OR work
Q.nodes().from_layers(L["social"] + L["work"])

# Difference: nodes in social but NOT bots
Q.nodes().from_layers(L["social"] - L["bots"])

# Intersection: nodes in both social AND work
Q.nodes().from_layers(L["social"] & L["work"])
```



```
# Complex: (social OR work) - bots
Q.nodes().from_layers(L["social"] + L["work"] - L["bots"])
```

This makes multilayer queries intuitive and expressive!

## Simple Layer Selection

```
# Single layer
Q.nodes().from_layers(L["social"])

# Multiple layers (union)
Q.nodes().from_layers(L["social"] + L["work"])
```

## Layer Algebra

**Union (OR):**

```
# Nodes in social OR work layers
Q.nodes().from_layers(L["social"] + L["work"])
```

**Difference (NOT):**

```
# Nodes in social but NOT in bots layer
Q.nodes().from_layers(L["social"] - L["bots"])
```

**Intersection (AND):**

```
# Nodes present in both social AND work
Q.nodes().from_layers(L["social"] & L["work"])
```

**All layers:**

```
# Don't specify from_layers() to query all layers
Q.nodes().where(degree__gt=5)
```

## Filtering Conditions

The `where()` method accepts filtering predicates.

## Degree Filters

```
# Greater than
Q.nodes().where(degree__gt=5)

# Greater than or equal
Q.nodes().where(degree__gte=3)

# Less than
Q.nodes().where(degree__lt=10)

# Less than or equal
```

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```
Q.nodes().where(degree__lte=5)
```

```
# Equal
```

```
Q.nodes().where(degree=2)
```

## Layer Filters

```
# Specific layer
```

```
Q.nodes().where(layer="social")
```

```
# Exclude layer
```

```
Q.nodes().where(layer__ne="bots")
```

## Multiple Conditions (AND)

```
# Combine conditions
```

```
Q.nodes().where(layer="social", degree__gt=3)
```

```
# Equivalent to: layer="social" AND degree > 3
```

## Computing Measures

The `compute()` method calculates network measures.

### Single Measure

```
result = (
    Q.nodes()
    .compute("degree")
    .execute(network)
)

# Access results
for node, degree in result.measures['degree'].items():
    print(f"{node}: {degree}")
```

### Multiple Measures

```
result = (
    Q.nodes()
    .compute("degree", "betweenness centrality", "closeness centrality")
    .execute(network)
)

# Access each measure
degrees = result.measures['degree']
betweenness = result.measures['betweenness centrality']
closeness = result.measures['closeness centrality']
```

## Available Measures

| Measure                | Description                                       |
|------------------------|---------------------------------------------------|
| degree                 | Node degree (number of neighbors)                 |
| degree_centrality      | Normalized degree centrality                      |
| betweenness_centrality | Betweenness centrality (path-based importance)    |
| closeness_centrality   | Closeness centrality (average distance to others) |
| eigenvector_centrality | Eigenvector centrality (importance of neighbors)  |
| pagerank               | PageRank score                                    |
| clustering             | Clustering coefficient                            |

## Ordering and Limiting

### Order By

```
# Order by degree (ascending)
Q.nodes().compute("degree").order_by("degree")

# Order by degree (descending)
Q.nodes().compute("degree").order_by("-degree")

# Order by multiple keys
Q.nodes().compute("degree").order_by("-degree", "node_id")
```

### Limit

```
# Get top 10 by degree
result = (
    Q.nodes()
    .compute("degree")
    .order_by("-degree")
    .limit(10)
    .execute(network)
)
```

## Working with Results

The `QueryResult` object provides multiple ways to access results.

### Iteration

```
result = Q.nodes().execute(network)

for node in result:
    print(node)  # ('Alice', 'friends'), ...
```

### Count

```
result = Q.nodes().where(degree__gt=5).execute(network)
print(f"Found {result.count} nodes")
```

### Measures

```
result = Q.nodes().compute("degree").execute(network)

# Access as dict
degrees = result.measures['degree']
for node, degree in degrees.items():
    print(f"{node}: {degree}")
```

### To Pandas

```
# Convert to DataFrame
df = result.to_pandas()
print(df.head())
```

### String DSL Syntax (Reference)

For completeness, here's the string DSL syntax:

#### Basic queries:

```
# Select nodes
execute_query(network, 'SELECT nodes')

# Filter by layer
execute_query(network, 'SELECT nodes WHERE layer="social"')

# Filter by degree
execute_query(network, 'SELECT nodes WHERE degree > 5')

# Multiple conditions
execute_query(network, 'SELECT nodes WHERE layer="social" AND degree > 3')

# Compute measures
execute_query(network, 'SELECT nodes COMPUTE betweenness centrality')
```

#### Operators:

- = — Equal
- != — Not equal
- > — Greater than
- < — Less than
- >= — Greater or equal
- <= — Less or equal
- AND — Logical AND

- OR — Logical OR
- NOT — Logical NOT

### Example: Complete Workflow

Here's a complete example demonstrating DSL capabilities:

```
from py3plex.core import multinet
from py3plex.dsl import Q, L

# Create network
network = multinet.multi_layer_network()
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Bob', 'friends', 'Carol', 'friends', 1],
    ['Carol', 'friends', 'Dave', 'friends', 1],
    ['Alice', 'colleagues', 'Bob', 'colleagues', 1],
    ['Bob', 'colleagues', 'Eve', 'colleagues', 1],
    ['Eve', 'colleagues', 'Frank', 'colleagues', 1],
], input_type="list")

# Query 1: Find high-degree nodes across all layers
result = (
    Q.nodes()
    .where(degree__gte=2)
    .compute("degree", "betweenness centrality")
    .order_by("-betweenness centrality")
    .execute(network)
)

print("High-degree nodes:")
for node in result:
    degree = result.measures['degree'][node]
    bc = result.measures['betweenness centrality'][node]
    print(f" {node}: degree={degree}, BC={bc:.3f}")

# Query 2: Compare layers
for layer in ['friends', 'colleagues']:
    result = (
        Q.nodes()
        .from_layers(L[layer])
        .compute("degree")
        .execute(network)
    )
    avg_degree = sum(result.measures['degree'].values()) / result.count
    print(f"{layer} layer: {result.count} nodes, avg degree={avg_degree:.2f}")
```

### Common Patterns

#### Pattern 1: Filter → Compute → Order → Limit

```
# Find top 10 central nodes in a specific layer
top_10 = (
```

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```
Q.nodes()
    .from_layers(L["social"])
    .compute("betweenness centrality")
    .order_by("-betweenness centrality")
    .limit(10)
    .execute(network)
)
```

## Pattern 2: Layer Comparison

```
# Compare degree distribution across layers
for layer_name in network.get_layers():
    result = (
        Q.nodes()
        .from_layers(L[layer_name])
        .compute("degree")
        .execute(network)
    )
    degrees = list(result.measures['degree'].values())
    print(f"{layer_name}: mean degree = {sum(degrees)/len(degrees):.2f}")
```

## Pattern 3: Conditional Export

```
# Export high-degree nodes to CSV
(
    Q.nodes()
    .where(degree__gt=5)
    .compute("degree", "betweenness centrality")
    .order_by("-degree")
    .export_csv("high_degree_nodes.csv")
    .execute(network)
)
```

## Summary

This chapter introduced the py3plex DSL:

1. **Motivation** — Declarative queries replace explicit loops
2. **Builder API** — Python-native with type hints (recommended)
3. **Query structure** — Target, layers, conditions, measures
4. **Layer algebra** — Union, difference, intersection operations
5. **Results** — Iteration, measures, pandas export

### Key takeaways:

- Use **Builder API** for type safety and autocompletion
- **Chain methods** to build complex queries incrementally
- **Layer algebra** enables powerful layer filtering
- **Compute measures** in a single call instead of manual loops

The next chapter covers the EXPLAIN mode for understanding query execution and performance.

### Further Reading

- The Builder API and Explain Plans
- Advanced Queries and Workflows
- `examples/network_analysis/example_dsl_builder_api.py` — Complete examples

## 1.7.9 The Builder API and Explain Plans

*TODO: Expand with examples from docfiles/user\_guide/dsl.rst sections on builder API*

### Builder Pattern and Fluent API

[Explain chainable method pattern]

### Method Chaining

[Examples of building queries incrementally]

### Type Hints and IDE Support

[Show autocompletion benefits]

### EXPLAIN Mode

#### Query Execution Plans

[Show how queries are parsed and planned]

```
from py3plex.dsl import Q

# Get execution plan without running query
plan = Q.nodes().where(degree__gt=5).explain()
print(plan)
```

### Complexity Estimates

[Understand query performance]

### Optimization Tips

[How to write efficient queries]

### Advanced Builder Features

#### Parameterized Queries

[Safe parameter binding]

```
from py3plex.dsl import Q, Param

# Parameterized query
result = (
```

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```
Q.nodes()
    .where(degree__gt=Param('min_degree'))
    .execute(network, params={'min_degree': 5})
)
```

## Query Reuse

[Build queries once, execute multiple times]

## Summary

[Key points about builder API and explain]

Source files: - docfiles/user\_guide/dsl.rst (builder API sections) - examples/network\_analysis/example\_dsl\_builder\_api.py

## 1.7.10 Advanced Queries and Workflows

This chapter covers advanced DSL patterns including dynamics simulation, complex query patterns, and workflow integration.

### DSL-Based Dynamics Simulation

The py3plex DSL includes a powerful framework for declarative dynamics simulation on multilayer networks. This section demonstrates how to use the dynamics DSL for epidemic modeling, random walks, and other dynamical processes.

### Mathematical Formalism

#### SIS Model (Susceptible-Infected-Susceptible)

The SIS model tracks disease spread with possible reinfection:

- **States:** S (susceptible), I (infected)
- **Update rules:**
  - $S \rightarrow I$  with probability  $= 1 - (1 - \beta)^{A \cdot I}$
  - $I \rightarrow S$  with probability  $\gamma$
- **Parameters:**
  - $\beta$ : transmission probability per contact (0 – 1)
  - $\gamma$ : recovery probability per time step (0 – 1)

The SIS model exhibits endemic equilibria when  $R_0 = \beta / \gamma \cdot \lambda_1(A) > 1$ , where  $\lambda_1(A)$  is the largest eigenvalue of the adjacency matrix.

#### SIR Model (Susceptible-Infected-Recovered)

The SIR model includes permanent immunity after recovery:

- **States:** S (susceptible), I (infected), R (recovered/removed)
- **Update rules:**
  - $S \rightarrow I$  with probability  $= 1 - (1 - \beta)^{A \cdot I}$
  - $I \rightarrow R$  with probability  $\gamma$



- **Parameters:**

- : transmission probability per contact
- : recovery probability per time step

The final outbreak size (attack rate) depends on the basic reproduction number  $R_0 = \beta \cdot \langle k \rangle$ , where  $\langle k \rangle$  is the mean degree.

### Random Walk

Random walk dynamics model diffusion processes:

- **State:** Current node position
- **Update rule:** Move to neighbor  $j$  with probability  $1/\text{degree}(i)$ , or stay at  $i$  with probability  $p_{\text{lazy}}$  (for lazy walks)
- **Parameters:**
  - teleport: probability of random teleportation (default: 0.05)

### Basic Dynamics DSL Usage

The dynamics DSL uses a builder API similar to the query DSL:

```
from py3plex.core import multinet
from py3plex.dynamics import D, SIS
from py3plex.dsl import L

# Create network
network = multinet.multi_layer_network()
# ... add nodes and edges ...

# Define SIS simulation
sim = (
    D.process(SIS(beta=0.3, mu=0.1)) # Specify process and parameters
    .initial(infected=0.05)          # 5% initially infected
    .steps(100)                      # Run for 100 time steps
    .measure("prevalence", "incidence") # Track measures
    .replicates(10)                  # Run 10 independent simulations
    .seed(42)                        # For reproducibility
)

# Execute simulation
result = sim.run(network)

# Access results
print(f"Mean final prevalence: {result.data['prevalence'][:, -1].mean():.3f}")

# Convert to pandas for analysis
df_dict = result.to_pandas()
prevalence_df = df_dict['prevalence']
```

#### Key components:

1. **D.process()** — Specify the dynamical process (SIS, SIR, RandomWalk)
2. **.initial()** — Set initial conditions
3. **.steps()** — Number of time steps to simulate

4. `.measure()` — Metrics to track during simulation
5. `.replicates()` — Number of independent runs (for averaging)
6. `.seed()` — Random seed for reproducibility
7. `.run()` — Execute the simulation

## Available Measures

Different processes support different measures:

| Process    | Measure             | Description                                  |
|------------|---------------------|----------------------------------------------|
| SIS        | prevalence          | Fraction of infected nodes at each time step |
| SIS        | incidence           | Number of new infections at each time step   |
| SIS        | prevalence_by_layer | Prevalence tracked separately per layer      |
| SIR        | prevalence          | Fraction of infected nodes                   |
| SIR        | state_counts        | Counts of nodes in each state (S, I, R)      |
| RandomWalk | visit_frequency     | Frequency of visits to each node             |
| RandomWalk | state_counts        | Current position over time                   |

## Multilayer Dynamics

The DSL supports multilayer networks with coupling between layers:

```
from py3plex.dsl import L

# Simulate on specific layers
sim = (
    D.process(SIS(beta=0.25, mu=0.08))
    .on_layers(L["offline"] + L["online"]) # Select layers
    .coupling(node_replicas="strong")      # Shared node states
    .initial(infected=0.1)
    .steps(120)
    .measure("prevalence", "prevalence_by_layer")
    .replicates(15)
)

result = sim.run(multilayer_network)
```

### Coupling modes:

- “strong” — Node states shared across all layers (default)
- “independent” — Each layer has independent node states
- “weak” — Partial coupling with cross-layer transmission rate

## Integration with Query DSL

The dynamics DSL integrates seamlessly with the query DSL for targeted initial conditions:

```
from py3plex.dsl import Q

# Start infection at high-degree nodes (hubs)
```

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```
sim = (
    D.process(SIS(beta=0.35, mu=0.12))
    .initial(
        infected=Q.nodes().where(degree__gte=5) # Query for hubs
    )
    .steps(100)
    .measure("prevalence")
    .replicates(10)
)

result = sim.run(network)
```

This allows precise control over initial conditions based on network structure, centrality, or other properties.

### Parameter Comparison Example

Compare dynamics under different parameters:

```
# Compare transmission rates
beta_values = [0.2, 0.3, 0.4, 0.5]
results = {}

for beta in beta_values:
    sim = (
        D.process(SIS(beta=beta, mu=0.1))
        .initial(infected=0.05)
        .steps(80)
        .measure("prevalence")
        .replicates(20)
        .seed(42)
    )
    results[beta] = sim.run(network)

# Analyze final prevalence
for beta, result in results.items():
    mean_final = result.data['prevalence'][:, -1].mean()
    print(f"{beta:.1f}: final prevalence = {mean_final:.3f}")
```

### Result Analysis

The `SimulationResult` object provides rich analysis capabilities:

```
# Get summary statistics
summary = result.summary()
print(summary)

# Plot time series with confidence intervals
import matplotlib.pyplot as plt
result.plot("prevalence")
plt.show()

# Export to pandas for custom analysis
```

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```
df_dict = result.to_pandas()
prevalence_df = df_dict['prevalence']

# Compute mean trajectory
mean_trajectory = (
    prevalence_df
    .groupby('t')['value']
    .agg(['mean', 'std'])
)
```

This chapter explores advanced DSL patterns for complex multilayer network analyses.

### Advanced DSL Patterns

The DSL supports sophisticated analysis workflows:

```
from py3plex.dsl import Q, L, Param
```

*# 1. Parameterized queries for systematic analysis*

```
query = (
    Q.nodes()
    .from_layers(L[Param.str("layer")])
    .where(degree__gt=Param.int("threshold"))
    .compute("betweenness centrality")
    .limit(Param.int("top_n"))
)

# Execute with different parameters
for layer in ["social", "work", "family"]:
    result = query.execute(network, layer=layer, threshold=5, top_n=20)
    result.to_pandas().to_csv(f"{layer}_hubs.csv")
```

*# 2. Multi-layer comparative analysis*

```
comparison = []
for layer in network.get_layers():
    stats = (
        Q.nodes()
        .from_layers(L[layer])
        .compute("degree", "betweenness centrality", "clustering")
        .execute(network)
    )
    df = stats.to_pandas()
    comparison.append({
        'layer': layer,
        'nodes': stats.count,
        'avg_degree': df['degree'].mean(),
        'max_bc': df['betweenness centrality'].max(),
    })
```

*# 3. EXPLAIN mode for optimization*

```
expensive_query = Q.nodes().compute("betweenness centrality")
plan = expensive_query.explain().execute(network)
for step in plan.steps:
```

```
print(f"{step.description}: {step.estimated_complexity}")
```

Advanced patterns enable complex research workflows!

*TODO: Expand from advanced DSL examples and patterns*

## Complex Query Patterns

### Multilayer Motifs

[Pattern detection across layers]

### Multilayer Paths

[Path queries respecting layer structure]

### Aggregations and Grouping

[Aggregate measures by layer]

### Result Conversion

#### To Pandas DataFrames

```
df = result.to_pandas()
```

#### To NetworkX

[Export query results as NetworkX graphs]

#### To Arrow/Parquet

[High-performance export]

### Workflow Integration

#### Pipeline Composition

[Chain multiple queries]

### Combining with sklearn

[Integration with machine learning pipelines]

### Custom Measures

[Extending the DSL with custom functions]

### Performance Tips

#### Query Optimization

For large-scale dynamics simulations:

- Use fewer replicates during development, scale up for production

- Consider using numpy backend for faster computation
- Track only essential measures to reduce memory usage
- Use smaller time steps when necessary but be aware of computational cost

### Memory Management

For very large networks or long simulations:

- Avoid tracking full trajectories (`trajectory` measure) if not needed
- Use measure-specific exports rather than full result objects
- Process results in batches for very long time series

### Summary

This chapter covered:

1. **DSL-based dynamics simulation** — Declarative framework for SIS, SIR, and Random Walk
2. **Mathematical formalism** — Formal definitions of epidemic models
3. **Multilayer coupling** — Simulating dynamics across network layers
4. **Query integration** — Using `Q.nodes()` for targeted initial conditions
5. **Parameter comparison** — Systematic exploration of parameter space
6. **Result analysis** — Rich result objects with pandas export and plotting

#### Key takeaways:

- The dynamics DSL follows the same design philosophy as the query DSL
- Simulations are fully declarative and composable
- Integration with query DSL enables sophisticated initial condition specification
- Results are analysis-ready with pandas, xarray, and plotting support

### Further Reading

- Introduction to the Py3plex DSL
- The Builder API and Explain Plans
- `examples/network_analysis/example_dsl_dynamics.py` — Complete examples
- `examples/advanced/example_dynamics_core.py` — Core dynamics classes
- `docfiles/sir_epidemic_simulator.rst` — SIR multiplex simulator documentation

## 1.7.11 Limitations and Stability Guarantees

*This chapter explicitly marks feature status for user confidence*

### DSL Feature Status

#### Stable APIs

Features with strong backward-compatibility guarantees:

- **Node queries** — `Q.nodes()` with filtering and measures

- **Layer algebra** — Union, difference, intersection operations
- **Core measures** — degree, betweenness, closeness, pagerank
- **Result export** — pandas, JSON, CSV formats

**Guarantee:** Stable APIs will not break in minor version updates ( $1.x \rightarrow 1.y$ ).

## Experimental Features

Functional but may change in future versions:

- **Edge queries** — `Q.edges()` (limited implementation)
- **Advanced aggregations** — `GROUP BY` equivalents
- **Nested subqueries** — Complex query composition

**Note:** Experimental features are marked in documentation. Use with awareness that API may evolve.

## Planned Features

Roadmap items (briefly mentioned, not detailed):

- **Full edge query support** — Complete parity with node queries
- **Graph pattern matching** — Motif detection DSL
- **Temporal queries** — Time-aware filtering

[Keep this section SHORT—no long speculative lists]

## Current Limitations

### Query Capabilities

What the DSL **can** do:

- Node filtering by degree, layer, and computed measures
- Compute centrality measures
- Layer algebra operations
- Export results in multiple formats

What the DSL **cannot** (yet) do:

- Complex edge queries (partial support only)
- Nested subqueries
- Aggregate functions (SUM, AVG, etc.)
- Join operations across layers

### Performance Boundaries

- **Small networks** (<10k nodes) — No issues
- **Medium networks** (10k-100k nodes) — Most queries fast
- **Large networks** (>100k nodes) — Some queries may be slow; use filtering early

### Scale and Performance

#### Query Complexity

[Table showing  $O(n)$ ,  $O(n \log n)$ ,  $O(n^2)$  operations]

#### Memory Requirements

[Guidelines for large networks]

#### When Not to Use the DSL

Use direct NetworkX/NumPy for:

- Single-layer graphs (DSL overhead not needed)
- Custom algorithms requiring fine-grained control
- Performance-critical inner loops

### API Versioning Policy

#### Semantic Versioning

py3plex follows semantic versioning (MAJOR.MINOR.PATCH):

- **MAJOR** (1.x → 2.x) — Breaking changes possible
- **MINOR** (1.1 → 1.2) — New features, stable API unchanged
- **PATCH** (1.1.0 → 1.1.1) — Bug fixes only

#### Deprecation Policy

Deprecated features are:

1. Marked in documentation and code (warnings)
2. Maintained for at least one minor version
3. Removed in next major version

### Migration Guides

[Commit to providing migration guides for breaking changes]

### Summary

#### Stable and production-ready:

- Node queries with filtering
- Core centrality measures
- Layer algebra
- Result export

#### Use with awareness:

- Experimental features may change
- Large network performance depends on query structure



- Some features are planned but not yet implemented

[Focus on transparency and user confidence]

Source files: - docfiles/user\_guide/dsl.rst (limitations sections) - docfiles/algorithm\_roadmap.rst (for planned features—keep brief)

## 1.7.12 Case Study 1 — Social Multiplex Network

### DSL in Case Studies

This case study demonstrates DSL throughout the analysis workflow:

```
from py3plex.dsl import Q, L

# 1. Quick exploratory analysis
for platform in ["facebook", "twitter", "linkedin"]:
    stats = (
        Q.nodes()
        .from_layers(L[platform])
        .compute("degree", "clustering")
        .execute(network)
    )
    df = stats.to_pandas()
    print(f"{platform}: avg degree = {df['degree'].mean():.2f}")

# 2. Find cross-platform influencers
influencers = (
    Q.nodes()
    .from_layers(L["facebook"] + L["twitter"] + L["linkedin"])
    .where(degree__gt=50)
    .compute("betweenness centrality")
    .order_by("-betweenness centrality")
    .limit(20)
    .export_csv("influencers.csv")
    .execute(network)
)

# 3. Platform-specific analysis
twitter_only = (
    Q.nodes()
    .from_layers(L["twitter"] - L["facebook"] - L["linkedin"])
    .compute("degree")
    .execute(network)
)
```

DSL enables rapid iteration in exploratory research!

*TODO: Develop from user\_guide/case\_studies.rst and examples*

### Domain Context

[Describe a real social network dataset — e.g., users across multiple platforms]

### Dataset: [Name TBD]

- **Nodes:** Users
- **Layers:** Facebook, Twitter, LinkedIn (or similar)
- **Size:** ~5,000 users, ~20,000 edges
- **Attributes:** User demographics, timestamps

### Data Structure

[How the data is structured as a multilayer network]

### Loading the Data

```
from py3plex.core import multinet

network = multinet.multi_layer_network(network_type='multiplex')
network.load_network('social_multiplex.edgelist', input_type='edgelist')

# Verify structure
network.basic_stats()
```

### Full Analysis Pipeline

#### Step 1: Exploratory Analysis

[Basic statistics, layer densities, node activities]

#### Step 2: Community Detection

[Identify communities across layers]

```
from py3plex.algorithms.community_detection import multilayer_louvain

communities = multilayer_louvain.best_partition(network.core_network)
```

#### Step 3: Centrality Analysis

[Find influential users using multilayer centrality]

#### Step 4: Cross-Layer Patterns

[Analyze how user roles vary across platforms]

#### Step 5: Visualization

[Publication-ready visualizations]

## Key Findings

[Scientific insights from the analysis]

## Community Structure

[What communities were found]

## Cross-Layer Roles

[How user importance varies by platform]

## Pitfalls Encountered

[Lessons learned]

## Data Cleaning Issues

[Real challenges with messy data]

## Performance Considerations

[What worked, what was slow]

## Reproducibility

[How to reproduce this analysis]

## Code Repository

[Link to example script]

## Data Availability

[Where to get the dataset]

## Summary

[Recap of workflow and findings]

*Source files:* - docfiles/user\_guide/case\_studies.rst - examples/ (select appropriate example) - Potential datasets: examples/datasets/ or multilayer\_datasets/

## 1.7.13 Case Study 2 — Biological Multilayer Network

*TODO: Different domain from Case Study 1—focus on dynamics/epidemic modeling*

### Domain Context

[Biological network — e.g., protein interactions or gene regulatory network]

### Dataset: [Name TBD]

- **Nodes:** Proteins or genes
- **Layers:** Different tissues, conditions, or interaction types

- **Size:** ~2,000 entities, ~10,000 interactions
- **Focus:** Dynamics and spreading processes

### Data Structure

[How biological data maps to multilayer structure]

### Loading and Preprocessing

```
network = multinet.multi_layer_network()  
# Load from biological database format
```

### Full Analysis Pipeline

#### Step 1: Network Topology

[Basic structural properties]

#### Step 2: Dynamics Simulation

[Model information spreading or cascade dynamics]

```
from py3plex.dynamics.models import SIRdynamics  
  
sir = SIRdynamics(network, beta=0.3, gamma=0.1)  
sir.set_seed(42)  
results = sir.run(steps=100)
```

#### Step 3: Layer-Specific Analysis

[How dynamics differ across layers]

#### Step 4: Intervention Analysis

[What-if scenarios: node removal, layer removal]

### Key Findings

[Biological insights]

### Spreading Patterns

[How processes propagate]

### Layer Interactions

[Cross-layer effects]

### Summary

[Workflow and insights]

Source files: - DYNAMICS\_IMPLEMENTATION.md - examples/ (dynamics examples) - doc-files/sir\_epidemic\_simulator.rst

### 1.7.14 Case Study 3 — Transportation Network (Optional)

*TODO: Advanced case study with heavy DSL usage or large-scale analysis*

[This chapter is optional and can be developed if time permits]

[Focus: Different from cases 1 & 2—perhaps temporal analysis or large-scale network]

#### Domain Context

[Transportation or infrastructure network]

#### Full Analysis Pipeline

[Complete workflow emphasizing advanced techniques]

#### Summary

[Insights and methodology]

*Source files:* - examples/ (advanced examples) - docfiles/user\_guide/case\_studies.rst

### 1.7.15 Testing and Validation

*TODO: Consolidate from docfiles/dev/ and tests/ structure*

#### Testing Philosophy

[High-level overview of py3plex’s testing approach]

#### Test Categories

- **Unit tests** — Individual functions and methods
- **Integration tests** — Module interactions
- **Property-based tests** — Hypothesis-driven testing
- **Regression tests** — Reference runs for dynamics

#### Test Organization

[Structure of tests/ directory]

```
tests/
├── test_core.py           # Core data structures
├── test_dsl*.py          # DSL functionality
├── test_dynamics*.py     # Dynamics models
├── test_algorithms*.py   # Algorithm correctness
└── property/             # Property-based tests
```

#### Key Validation Strategies

##### Random Walk Conservation

[Test that random walks conserve probability]

```
# Example validation pattern
def test_random_walk_conservation():
    """Ensure random walk probabilities sum to 1."""
    # Test code here
```

### Node2Vec Validation

[Bias checks, parameter sensitivity]

### Community Detection Validation

[Modularity bounds, partition quality]

### Dynamics Validation

[Conservation laws, steady-state properties]

## Running Tests

### Basic Test Execution

```
# Run all tests
pytest tests/

# Run specific test file
pytest tests/test_dsl.py

# Run with coverage
pytest tests/ --cov=py3plex
```

### Using Makefile

```
make test          # Run all tests
make test-coverage # Generate coverage report
```

## Continuous Integration

[Brief overview—detailed CI configs in Appendix]

### GitHub Actions

[Test matrix: Python versions, OS platforms]

## Summary

[Testing ensures correctness—but keep details in Appendix C]

Source files: - tests/ directory structure - docfiles/dev/development\_guide.rst - Detailed test scripts → Appendix C

## 1.7.16 Reproducible Environments

*TODO: Consolidate environment setup practices—defer details to Appendix*

### Reproducibility Principles

[Why reproducibility matters in network science]

### Environment Management

#### Python Virtual Environments

```
# venv
python3 -m venv py3plex-env
source py3plex-env/bin/activate
pip install py3plex

# conda
conda create -n py3plex python=3.10
conda activate py3plex
pip install py3plex
```

### Dependency Pinning

```
# Generate requirements.txt
pip freeze > requirements.txt

# Reproduce environment
pip install -r requirements.txt
```

### Docker Containers (Overview)

[High-level benefits—detailed configs in Appendix B]

#### Why Docker?

- **Isolation** — No system Python conflicts
- **Reproducibility** — Identical environment across machines
- **Portability** — Share complete environment

#### Basic Docker Usage

```
# Build image
docker build -t py3plex:latest .

# Run analysis
docker run --rm py3plex:latest python analysis.py
```

[For detailed Dockerfile, Compose configs → See Appendix B]

## Seed Management

[Ensuring deterministic results]

### Setting Seeds

```
# For dynamics
from py3plex.dynamics.models import SIRDynamics

sir = SIRDynamics(network, beta=0.3, gamma=0.1)
sir.set_seed(42) # Reproducible results

# For random graph generation
import random
random.seed(42)
```

## Recording Environments

### Environment Captures

```
# Save environment info
import platform
import py3plex

print(f"Python: {platform.python_version()}")
print(f"py3plex: {py3plex.__version__}")
print(f"Platform: {platform.platform()}")
```

## Version Control

[Git for code, data versioning strategies]

## Summary

### Key practices:

- Use virtual environments or containers
- Pin dependencies with requirements.txt
- Set seeds for stochastic processes
- Record environment details in papers/reports

[Detailed Docker configs and CI setup → Appendices]

Source files: - docfiles/deployment/cli\_and\_docker.rst (high-level) - Dockerfile and docker-compose.yml → Appendix B - docfiles/dev/development\_guide.rst

### 1.7.17 The Py3plex GUI (Overview)

*TODO: High-level GUI overview—detailed deployment in Appendix B*



## What is the Py3plex GUI?

[Web-based interface for interactive network exploration]

### Key Features

- **Network upload** — Load from files
- **Interactive visualization** — Explore structure visually
- **Query interface** — DSL queries via web UI
- **Analysis workflows** — Common operations without coding

### Architecture

#### Technology Stack

- **Backend:** FastAPI (Python)
- **Frontend:** SvelteKit (JavaScript)
- **Visualization:** D3.js, Plotly
- **API:** RESTful HTTP endpoints

[For detailed architecture → See Appendix or docfiles/gui/gui\_architecture.rst]

### Running Locally

#### Quickstart

```
# Install with GUI dependencies
pip install 'py3plex[gui]'

# Start development server
cd gui
python app.py
```

The GUI will be available at `http://localhost:5000`.

### Configuration

```
# gui/config.py
DEBUG = True
HOST = '0.0.0.0'
PORT = 5000
UPLOAD_FOLDER = './uploads'
```

### Key Workflows

#### Upload and Explore

1. Upload network file (edgelist, GraphML, JSON)
2. View statistics and layer information
3. Interactive visualization
4. Export results

### Query Interface

[Execute DSL queries via web form]

### Analysis Pipelines

[Pre-built workflows: community detection, centrality, etc.]

### Limitations and Safety

#### Security Considerations

**WARNING:** The GUI is designed for **local use only**, not for public internet exposure.

- No authentication by default
- No rate limiting
- No input sanitization for public use
- File uploads are not sandboxed

**For production:** See Appendix B for:

- Nginx reverse proxy configuration
- Authentication setup
- Security hardening checklist

#### Scalability Limits

- **Small networks** (<1k nodes) — Full interactivity
- **Medium networks** (1k-10k nodes) — Some features slow
- **Large networks** (>10k nodes) — Use CLI instead of GUI

#### When to Use the GUI vs CLI

**Use GUI for:**

- Exploratory analysis
- Learning py3plex features
- Presenting to non-programmers
- Quick prototyping

**Use CLI/library for:**

- Large networks
- Production workflows
- Reproducible analyses
- Performance-critical tasks

## Summary

The py3plex GUI provides:

- **Interactive exploration** without coding
- **Visualization** for understanding structure
- **Query interface** for the DSL
- **Local deployment** for individual researchers

**Key limitations:**

- Not hardened for public deployment
- Performance limited for large networks
- Advanced features require CLI/library

[For deployment details → Appendix B] [For API reference → docfiles/gui/gui\_api\_reference.rst]

*Source files:* - docfiles/gui/ (all files) - gui/ (application code) - Deployment details → Appendix B

## 1.7.18 Appendix A: Repository Layout and Scripts

This appendix provides a reference for the py3plex repository structure and maps book examples to actual scripts.

### Repository Structure

#### Top-Level Organization

```
py3plex/
├── py3plex/           # Main package
├── tests/             # Test suite
├── examples/          # 50+ example scripts
├── docfiles/           # Documentation source (Sphinx)
├── book/              # This book's source files
├── gui/               # Web GUI application
├── benchmarks/        # Performance benchmarks
├── datasets/          # Sample datasets
├── pyproject.toml      # Package configuration
├── Makefile           # Development automation
└── README.md          # Project overview
```

#### Main Package Structure

```
py3plex/
├── __init__.py
├── core/              # Data structures
│   ├── multinet.py    # multi_layer_network class
│   └── random_generators.py
├── algorithms/        # Analysis algorithms
│   ├── statistics/
│   ├── centrality/
│   ├── community_detection/
│   └── paths/
└── dynamics/          # Dynamical processes
```

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```

├── models.py          # SIS, SIR, SEIR
├── core.py
├── compartmental.py
├── dsl/               # Query language
│   ├── __init__.py
│   ├── builder.py     # Builder API
│   ├── executor.py
│   └── export.py
├── visualization/     # Plotting
├── io/               # I/O handlers
│   ├── readers.py
│   └── writers.py
├── cli.py             # Command-line interface
├── graph_ops.py       # Dplyr-style API
├── pipeline.py        # sklearn-style pipelines
└── workflows.py       # Config-driven workflows

```

## Examples Directory

The `examples/` directory contains 50+ working scripts organized by topic:

```

examples/
├── getting_started/   # Introductory examples
├── network_analysis/  # Core analysis examples
│   ├── example_dsl_builder_api.py    # Chapter 8-10
│   ├── example_dsl_queries.py
│   ├── example_community_detection.py # Chapter 7
│   └── example_centrality.py
├── dynamics/         # Epidemic and process models
├── visualization/    # Plotting examples
├── communities/      # Community detection
├── io_and_data/      # Data loading
├── cli/              # Command-line usage
├── workflows/        # Complete workflows
└── advanced/         # Advanced topics

```

## Mapping Book Chapters to Examples

### Chapter 4 (Installation and Getting Started)

- `examples/getting_started/quickstart.py`
- `examples/getting_started/first_network.py`

### Chapter 5 (Data Loading)

- `examples/io_and_data/load_edgelist.py`
- `examples/io_and_data/arrow_parquet_io.py`
- `examples/io_and_data/csv_loading.py`

## Chapter 6 (Visualization)

- `examples/visualization/basic_plot.py`
- `examples/visualization/hairball_plot.py`
- `examples/visualization/matrix_visualization.py`

## Chapter 7 (Core Algorithms)

- `examples/network_analysis/example_community_detection.py`
- `examples/network_analysis/example_centrality.py`
- `examples/network_analysis/example_explainable_centrality.py`
- `examples/dynamics/sir_epidemic.py`

## Chapters 8-10 (DSL)

- `examples/network_analysis/example_dsl_builder_api.py` — **Primary reference**
- `examples/network_analysis/example_dsl_queries.py`
- `examples/network_analysis/example_dsl_advanced.py`
- `examples/network_analysis/example_dsl_community_detection.py`

## Chapter 17 (GUI)

- `gui/app.py` — Main application
- `gui/README.md` — Setup instructions

## Key Scripts Reference

### Makefile Targets

The Makefile provides common development tasks:

```
make setup          # Create virtual environment
make dev-install    # Install with dev dependencies
make test           # Run test suite
make test-coverage  # Generate coverage report
make lint           # Run code quality checks
make format         # Format code with black
make docs           # Build documentation
make clean          # Clean build artifacts
```

### Development Scripts

```
scripts/
├── run_tests.py      # Test runner
├── generate_docs.py  # Documentation generator
└── benchmark.py     # Performance benchmarking
```

### Configuration Files

#### Package Configuration

**pyproject.toml** — Modern Python package configuration (PEP 518):

- Package metadata (name, version, authors)
- Dependencies and extras ([`infomap`], [`viz`], etc.)
- Build system configuration
- Tool configurations (black, ruff, mypy)

#### Testing Configuration

**pytest.ini** — Test framework configuration:

- Test discovery patterns
- Coverage settings
- Markers for test categories

#### Documentation Configuration

**docfiles/conf.py** — Sphinx documentation configuration:

- Extensions (autodoc, napoleon, mathjax)
- Theme settings (sphinx\_rtd\_theme)
- Build options

#### Docker Configuration

**Dockerfile** — Container image definition **docker-compose.yml** — Multi-service orchestration

[Detailed Docker configs in Appendix B]

### Data Files

#### Sample Datasets

```
datasets/  
├── karate_multiplex.edgelist  
├── example_biological.graphml  
└── README.md
```

#### External Datasets

The `multilayer_datasets/` directory contains links and scripts for downloading larger datasets used in case studies.

### Summary

**Key directories:**

- `py3plex/` — Main package code
- `tests/` — Test suite
- `examples/` — 50+ working examples mapped to book chapters

- `docfiles/` — Documentation source
- `gui/` — Web interface

**Development workflow:**

1. Clone repository
2. `make setup` to create environment
3. `make dev-install` to install dependencies
4. Modify code, run `make test`
5. Use `examples/` as reference

**Finding code:**

- Algorithms → `py3plex/algorithms/`
- DSL → `py3plex/dsl/`
- Examples → `examples/` (organized by topic)

## 1.7.19 Appendix B: Docker, Docker Compose, and Deployment

This appendix provides complete Docker configurations and deployment instructions for py3plex, including the GUI.

**Dockerfile Reference****Main Dockerfile**

The repository includes a Dockerfile for containerized execution:

```
# File: Dockerfile
FROM python:3.10-slim

WORKDIR /app

# Install system dependencies
RUN apt-get update && apt-get install -y \
    git \
    build-essential \
    && rm -rf /var/lib/apt/lists/*

# Copy requirements
COPY pyproject.toml .
COPY README.md .

# Install py3plex
RUN pip install --no-cache-dir -e .

# Copy application code
COPY py3plex/ ./py3plex/
COPY examples/ ./examples/

# Set up entry point
ENTRYPOINT ["python", "-m", "py3plex"]
CMD ["--help"]
```

### Building the Image

```
# Build image
docker build -t py3plex:latest .

# Build with specific Python version
docker build --build-arg PYTHON_VERSION=3.11 -t py3plex:3.11 .

# Build with tag
docker build -t py3plex:1.0 .
```

### Running Containers

```
# Run command
docker run --rm py3plex:latest --version

# Run analysis script
docker run --rm py3plex:latest python examples/getting_started/quickstart.py

# Mount data directory
docker run --rm -v $(pwd)/data:/data py3plex:latest python analysis.py

# Interactive shell
docker run --rm -it py3plex:latest /bin/bash
```

### Docker Compose

#### GUI Deployment

```
# File: docker-compose.yml
version: '3.8'

services:
  gui:
    build:
      context: .
      dockerfile: gui/Dockerfile
    ports:
      - "5000:5000"
    volumes:
      - ./data:/app/data
      - ./uploads:/app/uploads
    environment:
      - FLASK_ENV=production
      - SECRET_KEY=${SECRET_KEY}
    restart: unless-stopped

# Optional: nginx reverse proxy
nginx:
  image: nginx:alpine
  ports:
    - "80:80"
```

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```

- "443:443"
volumes:
- ./nginx.conf:/etc/nginx/nginx.conf:ro
- ./ssl:/etc/nginx/ssl:ro
depends_on:
- gui
restart: unless-stopped

```

## Running with Docker Compose

```

# Start services
docker compose up -d

# View logs
docker compose logs -f gui

# Stop services
docker compose down

# Rebuild after changes
docker compose up -d --build

```

## Production Deployment

### Security Hardening

#### 1. Use environment variables for secrets:

```

# .env file (DO NOT commit to git)
SECRET_KEY=your-strong-secret-key-here
DATABASE_URL=postgresql://user:pass@host/db

```

```

# docker-compose.yml
services:
  gui:
    env_file:
      - .env

```

#### 2. Run as non-root user:

```

# Add to Dockerfile
RUN useradd -m -u 1000 py3plex
USER py3plex

```

#### 3. Use read-only file systems where possible:

```

services:
  gui:
    read_only: true
    tmpfs:
      - /tmp
      - /var/tmp

```

## Nginx Reverse Proxy

nginx.conf:

```
events {
    worker_connections 1024;
}

http {
    upstream gui {
        server gui:5000;
    }

    server {
        listen 80;
        server_name yourdomain.com;

        # Redirect to HTTPS
        return 301 https://$server_name$request_uri;
    }

    server {
        listen 443 ssl;
        server_name yourdomain.com;

        ssl_certificate /etc/nginx/ssl/cert.pem;
        ssl_certificate_key /etc/nginx/ssl/key.pem;

        # Security headers
        add_header X-Frame-Options "SAMEORIGIN";
        add_header X-Content-Type-Options "nosniff";
        add_header X-XSS-Protection "1; mode=block";

        location / {
            proxy_pass http://gui;
            proxy_set_header Host $host;
            proxy_set_header X-Real-IP $remote_addr;
            proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
            proxy_set_header X-Forwarded-Proto $scheme;
        }

        # File upload limits
        client_max_body_size 100M;
    }
}
```

## TLS/SSL Configuration

Using Let's Encrypt:

```
# Install certbot
docker run --rm -v $(pwd)/ssl:/etc/letsencrypt \
    certbot/certbot certonly --standalone \
```

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```
-d yourdomain.com \
--email your@email.com \
--agree-tos
```

### Self-signed certificates (development only):

```
# Generate self-signed certificate
openssl req -x509 -newkey rsa:4096 \
-keyout ssl/key.pem \
-out ssl/cert.pem \
-days 365 -nodes
```

## Authentication Setup

### Basic HTTP authentication (simple):

```
server {
    ...

    location / {
        auth_basic "Restricted Access";
        auth_basic_user_file /etc/nginx/.htpasswd;

        proxy_pass http://gui;
    }
}
```

```
# Generate .htpasswd file
htpasswd -c .htpasswd username
```

### OAuth2 Proxy (advanced):

Use oauth2-proxy as an additional service for integration with GitHub, Google, etc.

## Security Checklist

### Before Public Deployment

- ☐ Use HTTPS (TLS/SSL) for all connections
- ☐ Set strong SECRET\_KEY environment variable
- ☐ Enable authentication (HTTP Basic, OAuth2, etc.)
- ☐ Run containers as non-root user
- ☐ Use read-only file systems where possible
- ☐ Set appropriate file upload limits
- ☐ Enable security headers in nginx
- ☐ Regularly update base images (docker pull python:3.10-slim)
- ☐ Monitor logs for suspicious activity
- ☐ Implement rate limiting (via nginx or application)

- [ ] Sanitize user inputs (especially file uploads)
- [ ] Restrict file types for uploads
- [ ] Scan uploaded files for malware
- [ ] Use network isolation (Docker networks)
- [ ] Keep dependencies up to date (`pip install --upgrade`)

### Monitoring

```
# Add health checks to docker-compose.yml
services:
  gui:
    healthcheck:
      test: ["CMD", "curl", "-f", "http://localhost:5000/health"]
      interval: 30s
      timeout: 10s
      retries: 3
```

### Cloud Deployment

#### AWS Deployment

##### Using ECS (Elastic Container Service):

1. Push image to ECR (Elastic Container Registry)
2. Create ECS task definition
3. Deploy to ECS cluster
4. Use ALB (Application Load Balancer) for routing

##### Using EC2:

1. Launch EC2 instance
2. Install Docker
3. Copy docker-compose.yml to instance
4. Run `docker compose up -d`

#### Azure Deployment

##### Using Azure Container Instances:

```
az container create \
  --resource-group myResourceGroup \
  --name py3plex-gui \
  --image py3plex:latest \
  --cpu 2 --memory 4 \
  --ports 5000
```

## GCP Deployment

### Using Cloud Run:

1. Push image to Google Container Registry
2. Deploy to Cloud Run
3. Configure domain and SSL

## Performance Tuning

### Container Resource Limits

```
services:
  gui:
    deploy:
      resources:
        limits:
          cpus: '2.0'
          memory: 4G
        reservations:
          cpus: '1.0'
          memory: 2G
```

## Caching

Use Docker layer caching to speed up builds:

```
# Install dependencies first (cached layer)
COPY pyproject.toml .
RUN pip install -e .

# Copy code second (changes frequently)
COPY py3plex/ ./py3plex/
```

## Multi-Stage Builds

For smaller production images:

```
# Build stage
FROM python:3.10 as builder
WORKDIR /app
COPY pyproject.toml .
RUN pip install --user -e .

# Runtime stage
FROM python:3.10-slim
WORKDIR /app
COPY --from=builder /root/.local /root/.local
COPY py3plex/ ./py3plex/
ENV PATH=/root/.local/bin:$PATH
```

## Troubleshooting

### Common Issues

#### Issue: Container exits immediately

Check logs:

```
docker logs <container_id>
```

#### Issue: Permission denied on mounted volumes

Fix permissions:

```
chmod -R 777 ./data
```

Or run as correct user ID.

#### Issue: Out of memory

Increase Docker memory limits or container resources.

## Summary

This appendix provided:

- Complete Dockerfile and docker-compose.yml examples
- Security hardening guidelines
- Production deployment configurations (nginx, TLS, authentication)
- Cloud deployment options (AWS, Azure, GCP)
- Performance tuning strategies

#### Key recommendations for production:

1. Always use HTTPS
2. Enable authentication
3. Run as non-root
4. Monitor and log
5. Keep dependencies updated
6. Follow security checklist

[For high-level GUI overview → Chapter 17]

## 1.7.20 Appendix C: Detailed Validation Scripts

This appendix provides detailed validation strategies and test scripts for ensuring correctness of multilayer network algorithms. These complement the high-level overview in Chapter 15.

### Validation Philosophy

py3plex uses multiple validation strategies:

1. **Unit tests** — Test individual functions in isolation
2. **Property-based tests** — Test invariants that should always hold
3. **Reference runs** — Compare against known-good results

4. **Conservation tests** — Verify physical/mathematical laws (e.g., probability conservation)

## Random Walk Validation

### Conservation Tests

Random walks must conserve probability—the sum of transition probabilities from any state must equal 1:

```
# File: tests/test_random_walk_conservation.py
import pytest
import numpy as np
from py3plex.algorithms.paths import random_walk

def test_probability_conservation():
    """Verify random walk conserves probability."""
    network = create_test_network()

    # Run random walk
    walks = random_walk(network, num_walks=1000, walk_length=100)

    # Compute transition matrix
    trans_matrix = compute_transition_matrix(walks)

    # Check row sums = 1
    row_sums = trans_matrix.sum(axis=1)
    assert np.allclose(row_sums, 1.0, atol=1e-6)
```

### Bias Validation

Node2Vec introduces biases (return parameter  $p$ , in-out parameter  $q$ ). Validate that biases have the intended effect:

```
# File: tests/test_node2vec_bias.py
def test_return_bias():
    """Higher p should reduce probability of returning to previous node."""
    network = create_test_network()

    # Low p (encourage return)
    walks_low_p = node2vec_walk(network, p=0.25, q=1.0, num_walks=1000)
    return_rate_low = count_immediate_returns(walks_low_p)

    # High p (discourage return)
    walks_high_p = node2vec_walk(network, p=4.0, q=1.0, num_walks=1000)
    return_rate_high = count_immediate_returns(walks_high_p)

    assert return_rate_high < return_rate_low
```

### Reference Runs

Store reference outputs for deterministic tests:

```
# File: tests/test_dynamics_reference_runs.py
REFERENCE_DATA = {
    'sir_small_graph_42': {
```

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```

        'beta': 0.3,
        'gamma': 0.1,
        'steps': 100,
        'seed': 42,
        'expected_final_recovered': 0.75
    }
}

def test_sir_matches_reference():
    """Ensure SIR dynamics match reference run."""
    ref = REFERENCE_DATA['sir_small_graph_42']

    network = make_tiny_chain_graph()
    sir = SIRDynamics(network, beta=ref['beta'], gamma=ref['gamma'])
    sir.set_seed(ref['seed'])

    results = sir.run(steps=ref['steps'])
    final_recovered = results.get_measure('state_counts')[2][-1] / network.number_of_
    ↪ nodes()

    assert abs(final_recovered - ref['expected_final_recovered']) < 0.01

```

## Community Detection Validation

### Modularity Bounds

Modularity Q must be in range [-0.5, 1.0]:

```

# File: tests/test_community_bounds.py
def test_modularity_bounds():
    """Modularity must be in valid range."""
    network = create_test_network()

    communities = multilayer_louvain.best_partition(network.core_network)
    Q = compute_modularity(network, communities)

    assert -0.5 <= Q <= 1.0

```

### Partition Quality

Every node must be assigned to exactly one community:

```

def test_partition_completeness():
    """Every node must be in exactly one community."""
    network = create_test_network()
    communities = multilayer_louvain.best_partition(network.core_network)

    # All nodes accounted for
    assert set(communities.keys()) == set(network.get_nodes())

    # No overlapping assignments (in non-overlapping methods)
    assert len(communities) == len(set(communities.values()))

```



## Singleton Detection

Validate that isolated nodes form their own communities:

```
def test_singleton_communities():
    """Isolated nodes should form singleton communities."""
    network = create_network_with_isolates()
    communities = multilayer_louvain.best_partition(network.core_network)

    isolates = [n for n in network.get_nodes() if network.core_network.degree(n) == 0]

    # Each isolate in its own community
    for node in isolates:
        community = communities[node]
        same_community = [n for n, c in communities.items() if c == community]
        assert same_community == [node]
```

## Centrality Validation

### Degree Centrality

Degree centrality should match actual degree counts:

```
# File: tests/test_centrality_correctness.py
def test_degree_centrality():
    """Degree centrality should match computed degrees."""
    network = create_test_network()

    import networkx as nx
    degree_cent = nx.degree_centrality(network.core_network)

    for node in network.get_nodes():
        computed = degree_cent[node]
        expected = network.core_network.degree(node) / (network.number_of_nodes() - 1)
        assert abs(computed - expected) < 1e-10
```

### Betweenness Centrality

Validate using known structures (e.g., star graphs):

```
def test_betweenness_star_graph():
    """In a star graph, center has betweenness 1.0."""
    network = create_star_graph(n=5) # 1 center + 4 leaves

    import networkx as nx
    bc = nx.betweenness_centrality(network.core_network, normalized=True)

    center = ('center', 'layer')
    assert bc[center] == 1.0 # All paths go through center

    leaves = [n for n in network.get_nodes() if n != center]
    for leaf in leaves:
        assert bc[leaf] == 0.0 # Leaves are not on any shortest paths
```

## Dynamics Validation

### SIS Model Conservation

In SIS model,  $S(t) + I(t) = N$  for all  $t$ :

```
# File: tests/test_dynamics_conservation.py
def test_sis_conservation():
    """SIS model must conserve population."""
    network = create_test_network()
    N = network.number_of_nodes()

    sis = SISDynamics(network, beta=0.3, gamma=0.1)
    sis.set_seed(42)

    results = sis.run(steps=100)

    for t in range(100):
        S_t = results.get_measure('state_counts')[0][t]
        I_t = results.get_measure('state_counts')[1][t]
        assert S_t + I_t == N
```

### SIR Model Conservation

In SIR model,  $S(t) + I(t) + R(t) = N$  for all  $t$ :

```
def test_sir_conservation():
    """SIR model must conserve population."""
    network = create_test_network()
    N = network.number_of_nodes()

    sir = SIRDynamics(network, beta=0.3, gamma=0.1)
    sir.set_seed(42)

    results = sir.run(steps=100)

    for t in range(100):
        S_t = results.get_measure('state_counts')[0][t]
        I_t = results.get_measure('state_counts')[1][t]
        R_t = results.get_measure('state_counts')[2][t]
        assert S_t + I_t + R_t == N
```

## Steady State Validation

SIS model should reach steady state:

```
def test_sis_steady_state():
    """SIS should reach equilibrium."""
    network = create_test_network()

    sis = SISDynamics(network, beta=0.3, gamma=0.1)
    sis.set_seed(42)

    results = sis.run(steps=1000)
```

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```
prevalence = results.get_measure('prevalence')

# Check last 100 steps are stable
last_100 = prevalence[-100:]
variance = np.var(last_100)
assert variance < 0.01 # Low variance = steady state
```

## Property-Based Testing

### Using Hypothesis

Property-based tests use the hypothesis library to generate random inputs:

```
# File: tests/property/test_paths_properties.py
from hypothesis import given, strategies as st
from hypothesis import assume

@given(
    num_nodes=st.integers(min_value=3, max_value=20),
    num_layers=st.integers(min_value=1, max_value=5),
    seed=st.integers(min_value=0, max_value=1000)
)
def test_shortest_path_length_property(num_nodes, num_layers, seed):
    """Shortest path length should be <= graph diameter."""
    network = generate_random_network(num_nodes, num_layers, seed)
    assume(nx.is_connected(network.core_network)) # Skip disconnected graphs

    # Pick random source and target
    nodes = list(network.get_nodes())
    source, target = random.sample(nodes, 2)

    # Compute shortest path
    path = nx.shortest_path(network.core_network, source, target)
    path_length = len(path) - 1

    # Path length should be <= diameter
    diameter = nx.diameter(network.core_network)
    assert path_length <= diameter
```

## Multilayer-Specific Properties

```
@given(num_nodes=st.integers(min_value=2, max_value=10))
def test_node_activity_bounds(num_nodes):
    """Node activity must be in [0, 1]."""
    network = generate_multiplex_network(num_nodes)

    for node_id in range(num_nodes):
        activity = mls.node_activity(network, str(node_id))
        assert 0.0 <= activity <= 1.0
```

## Running Validation Tests

### Full Test Suite

```
# Run all tests
pytest tests/

# Run only validation tests
pytest tests/ -k validation

# Run with coverage
pytest tests/ --cov=py3plex --cov-report=html
```

### Property-Based Tests

```
# Run property tests (slower)
pytest tests/property/

# Run with more examples
pytest tests/property/ --hypothesis-profile=thorough
```

### Continuous Validation

Tests run automatically on:

- Every commit (GitHub Actions)
- Pull requests
- Scheduled nightly runs

### Summary

This appendix detailed validation strategies:

1. **Conservation tests** — Physical/mathematical laws must hold
2. **Reference runs** — Deterministic tests against known-good outputs
3. **Boundary tests** — Values must be in valid ranges
4. **Property-based tests** — Invariants hold for random inputs
5. **Structure tests** — Algorithmic correctness on known structures

#### Key test files:

- tests/test\_dynamics\_reference\_runs.py — Reference dynamics data
- tests/test\_random\_walk\_conservation.py — Probability conservation
- tests/property/ — Property-based tests
- tests/test\_centrality\_correctness.py — Centrality validation

[For high-level testing overview → Chapter 15]

## 1.7.21 Appendix D: Error Handling and Exception Hierarchy

This appendix documents py3plex's exception classes and error handling conventions.

### Exception Hierarchy

py3plex defines a hierarchy of exceptions for different error categories:

```
Py3plexException (base)
├── ValidationError
│   ├── InvalidNetworkError
│   ├── InvalidLayerError
│   └── InvalidNodeError
├── IOError
│   ├── FileFormatError
│   ├── FileNotFoundError
│   └── SerializationError
├── DSLError
│   ├── QuerySyntaxError
│   ├── QueryExecutionError
│   └── UnsupportedFeatureError
├── AlgorithmError
│   ├── ConvergenceError
│   └── InsufficientDataError
└── ConfigurationError
```

### Base Exception

```
class Py3plexException(Exception):
    """Base exception for all py3plex errors."""
    pass
```

All py3plex exceptions inherit from this base class, allowing users to catch any library error:

```
from py3plex.exceptions import Py3plexException

try:
    result = some_py3plex_operation()
except Py3plexException as e:
    print(f"py3plex error: {e}")
```

### Validation Errors

#### InvalidNetworkError

Raised when a network is in an invalid state:

```
from py3plex.exceptions import InvalidNetworkError

# Example: Empty network
if network.number_of_nodes() == 0:
    raise InvalidNetworkError("Cannot compute metrics on empty network")
```

Common causes:

- Empty network (no nodes or edges)
- Disconnected graph when algorithm requires connected
- Directed graph when undirected expected (or vice versa)

### InvalidLayerError

Raised when referencing a nonexistent layer:

```
from py3plex.exceptions import InvalidLayerError

if layer_name not in network.get_layers():
    raise InvalidLayerError(f"Layer '{layer_name}' does not exist")
```

#### Common causes:

- Typo in layer name
- Layer removed after reference created
- Case sensitivity mismatch

### InvalidNodeError

Raised when referencing a nonexistent node:

```
from py3plex.exceptions import InvalidNodeError

if node not in network.core_network:
    raise InvalidNodeError(f"Node {node} not found in network")
```

## I/O Errors

### FileFormatError

Raised when file format is invalid or unsupported:

```
from py3plex.exceptions import FileFormatError

try:
    network.load_network("data.xyz", input_type="xyz")
except FileFormatError as e:
    print(f"Unsupported format: {e}")
```

#### Common causes:

- Unrecognized file extension
- Malformed file content
- Missing required fields in CSV/JSON

### SerializationError

Raised during read/write operations:

```
from py3plex.exceptions import SerializationError

try:
    write(network, "output.arrow")
except SerializationError as e:
    print(f"Failed to serialize: {e}")
```

## DSL Errors

### QuerySyntaxError

Raised for invalid DSL query syntax:

```
from py3plex.dsl import execute_query
from py3plex.exceptions import QuerySyntaxError

try:
    result = execute_query(network, 'SELECT nodes WHRE degree > 5') # Typo
except QuerySyntaxError as e:
    print(f"Query syntax error: {e}")
    # Suggestion: "Did you mean 'WHERE'?"
```

#### Common causes:

- Typos in keywords (WHRE instead of WHERE)
- Missing quotes around layer names
- Invalid comparison operators

### QueryExecutionError

Raised when query execution fails:

```
from py3plex.exceptions import QueryExecutionError

try:
    result = execute_query(network, 'SELECT nodes COMPUTE nonexistent_measure')
except QueryExecutionError as e:
    print(f"Query execution failed: {e}")
```

#### Common causes:

- Referencing nonexistent measures
- Applying measure to unsuitable graph (e.g., eigenvector centrality on disconnected graph)
- Resource limits exceeded

### UnsupportedFeatureError

Raised when using experimental or unimplemented features:

```
from py3plex.exceptions import UnsupportedFeatureError

try:
```

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```
result = Q.edges().where(weight__gt=0.5).execute(network)
except UnsupportedFeatureError as e:
    print(f"Feature not yet supported: {e}")
```

## Algorithm Errors

### ConvergenceError

Raised when iterative algorithms fail to converge:

```
from py3plex.exceptions import ConvergenceError

try:
    centrality = compute_eigenvector_centrality(network, max_iter=100)
except ConvergenceError as e:
    print(f"Algorithm did not converge: {e}")
```

### InsufficientDataError

Raised when there's not enough data for analysis:

```
from py3plex.exceptions import InsufficientDataError

if network.number_of_nodes() < 3:
    raise InsufficientDataError("Need at least 3 nodes for community detection")
```

## Configuration Errors

### ConfigurationError

Raised for invalid configuration or parameters:

```
from py3plex.exceptions import ConfigurationError

if beta <= 0 or gamma <= 0:
    raise ConfigurationError("SIR parameters beta and gamma must be positive")
```

## Error Handling Best Practices

### Catch Specific Exceptions

Prefer catching specific exceptions over generic ones:

```
# Good: Specific exception handling
from py3plex.exceptions import InvalidLayerError

try:
    result = Q.nodes().from_layers(L["social"]).execute(network)
except InvalidLayerError:
    print("Layer 'social' not found. Available layers:", network.get_layers())

# Avoid: Generic exception (too broad)
```

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```
try:
    result = Q.nodes().from_layers(L["social"]).execute(network)
except Exception as e: # Too generic
    print(f"Something went wrong: {e}")
```

## Provide Context

Include helpful context in error messages:

```
# Good: Helpful context
if layer_name not in network.get_layers():
    available = ", ".join(network.get_layers())
    raise InvalidLayerError(
        f"Layer '{layer_name}' not found. Available layers: {available}"
    )

# Less helpful
if layer_name not in network.get_layers():
    raise InvalidLayerError(f"Invalid layer")
```

## Validation at Entry Points

Validate inputs early:

```
def compute_centrality(network, measure='degree'):
    """Compute node centrality."""
    # Validate early
    if network.number_of_nodes() == 0:
        raise InvalidNetworkError("Cannot compute centrality on empty network")

    if measure not in ['degree', 'betweenness', 'closeness']:
        raise ValueError(f"Unknown measure: {measure}")

    # Proceed with computation
    ...
```

## Clean Resource Cleanup

Use context managers for resources:

```
# Good: Automatic cleanup
from py3plex.io import read

try:
    graph = read('network.arrow')
    # Process graph
except IOError as e:
    print(f"Failed to read file: {e}")
finally:
    # Resources automatically cleaned up
```

### Error Messages Guidelines

1. **Be specific:** State what went wrong and why
2. **Suggest fixes:** Offer alternatives when possible
3. **Include context:** Show relevant values (layer names, node counts, etc.)
4. **Be concise:** One sentence explanation, then details if needed

#### Examples:

```
# Good error message
raise InvalidLayerError(
    f"Layer 'socail' not found. Did you mean 'social'? "
    f"Available layers: {'', ' '.join(network.get_layers())}"
)

# Good error message
raise InsufficientDataError(
    f"Community detection requires at least 3 nodes, but network has only "
    f"{network.number_of_nodes()}"
)

# Less helpful
raise InvalidLayerError("Layer error") # Too vague
```

### Logging

py3plex uses Python's logging module:

```
import logging

# Configure logging level
logging.basicConfig(level=logging.INFO)

# py3plex will log warnings and errors
network.load_network("data.edgelist") # Logs any warnings
```

Available log levels:

- DEBUG — Detailed information for debugging
- INFO — General information
- WARNING — Recoverable issues
- ERROR — Serious problems
- CRITICAL — Fatal errors

### Summary

#### Exception hierarchy:

- Base: Py3plexException
- Categories: Validation, I/O, DSL, Algorithm, Configuration

#### Best practices:

1. Catch specific exceptions
2. Provide helpful context in messages
3. Validate early
4. Suggest fixes when possible
5. Use logging for non-fatal issues

**Example usage:**

```
from py3plex.core import multinet
from py3plex.exceptions import InvalidLayerError, Py3plexException

try:
    network = multinet.multi_layer_network()
    network.load_network("data.edgelist")

    result = Q.nodes().from_layers(L["social"]).execute(network)

except InvalidLayerError as e:
    print(f"Layer not found: {e}")
except Py3plexException as e:
    print(f"py3plex error: {e}")
```

[For error recovery strategies in workflows → Chapter 10]

## 1.7.22 Appendix E: Extended API and DSL Reference

This appendix provides a quick reference for the py3plex API and DSL syntax.

### Core API Reference

#### Creating Networks

```
from py3plex.core import multinet

# Create empty network
network = multinet.multi_layer_network(
    directed=False,          # True for directed networks
    network_type='multilayer' # or 'multiplex'
)
```

#### Adding Nodes and Edges

```
# Add edges (nodes created implicitly)
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1.0],
    ['Bob', 'friends', 'Carol', 'friends', 1.0],
], input_type="list")

# Add edges with attributes
network.add_edges([
    {'source': 'Alice',
```

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```
'source_type': 'friends',  
'target': 'Bob',  
'target_type': 'friends',  
'weight': 0.8,  
'timestamp': '2023-01-15'  
}})
```

## Loading Networks

```
# From file  
network.load_network("data.edgelist", input_type="edgelist")  
  
# From NetworkX  
import networkx as nx  
G = nx.karate_club_graph()  
network.load_network_from_networkx(G, layer_name='social')
```

## Basic Operations

```
# Get network information  
layers = network.get_layers()  
nodes = list(network.get_nodes())  
edges = list(network.get_edges())  
  
# Statistics  
network.basic_stats()  
num_nodes = network.get_number_of_nodes()  
num_edges = network.get_number_of_edges()  
  
# Layer-specific  
nodes_per_layer = network.get_number_of_nodes_per_layer()  
layer_subgraph = network.get_layer_subgraph('friends')
```

## Algorithms API

### Statistics

```
from py3plex.algorithms.statistics import multilayer_statistics as mls  
  
# Layer density  
density = mls.layer_density(network, 'friends')  
  
# Node activity (fraction of layers a node appears in)  
activity = mls.node_activity(network, 'Alice')  
  
# Layer overlap  
overlap = mls.layer_overlap(network, 'friends', 'colleagues')
```

## Community Detection

```
from py3plex.algorithms.community_detection import multilayer_louvain

# Detect communities
communities = multilayer_louvain.best_partition(network.core_network)

# communities = {'Alice', 'friends': 0, ('Bob', 'friends'): 0, ...}
```

## Centrality

```
import networkx as nx

# Standard centralities (use NetworkX)
G = network.core_network
degree_cent = nx.degree_centrality(G)
betweenness = nx.betweenness_centrality(G)
closeness = nx.closeness_centrality(G)

# Explainable centrality (py3plex-specific)
from py3plex.algorithms.centrality.explain import explain_node_centrality

explanation = explain_node_centrality(
    network,
    node='Alice',
    measure='degree'
)
# Returns: {'score': 4, 'layer_breakdown': {...}, ...}
```

## Dynamics

```
from py3plex.dynamics.models import SIRDynamics, SISDynamics

# Configure SIR model
sir = SIRDynamics(
    network,
    beta=0.3,      # Infection rate
    gamma=0.1,     # Recovery rate
    initial_infected={'Alice': 0} # Initial conditions
)

# Set seed for reproducibility
sir.set_seed(42)

# Run simulation
results = sir.run(steps=100)

# Extract measures
prevalence = results.get_measure('prevalence')
recovered = results.get_measure('state_counts')[2] # State 2 = recovered
```

## DSL Quick Reference

### Builder API (Recommended)

```

from py3plex.dsl import Q, L, Param

# Basic query
result = Q.nodes().execute(network)

# Filter by layer
result = Q.nodes().from_layers(L["social"]).execute(network)

# Filter by condition
result = Q.nodes().where(degree__gt=5).execute(network)

# Multiple conditions
result = Q.nodes().where(layer="social", degree__gt=3).execute(network)

# Layer algebra
result = Q.nodes().from_layers(L["social"] + L["work"]).execute(network)      # Union
result = Q.nodes().from_layers(L["social"] - L["bots"]).execute(network)      # Difference
result = Q.nodes().from_layers(L["social"] & L["work"]).execute(network)      # Intersection
↪ Intersection

# Compute measures
result = Q.nodes().compute("degree", "betweenness centrality").execute(network)

# Order and limit
result = (
    Q.nodes()
    .compute("degree")
    .order_by("-degree")
    .limit(10)
    .execute(network)
)

# Export results
(
    Q.nodes()
    .compute("degree")
    .export_csv("output.csv")
    .execute(network)
)

# Parameterized queries
result = (
    Q.nodes()
    .where(degree__gt=Param('min_degree'))
    .execute(network, params={'min_degree': 5})
)

```

## String DSL Syntax

```
from py3plex.dsl import execute_query

# Basic syntax: SELECT target WHERE conditions COMPUTE measures

# Select all nodes
execute_query(network, 'SELECT nodes')

# Filter by layer
execute_query(network, 'SELECT nodes WHERE layer="social"')

# Filter by degree
execute_query(network, 'SELECT nodes WHERE degree > 5')

# Multiple conditions
execute_query(network, 'SELECT nodes WHERE layer="social" AND degree > 3')

# Compute measures
execute_query(network, 'SELECT nodes COMPUTE betweenness centrality')

# Combined
execute_query(network,
    'SELECT nodes WHERE layer="social" AND degree > 3 '
    'COMPUTE degree betweenness centrality'
)
```

## DSL Operators

### Comparison operators:

- `=` — Equal to
- `!=` — Not equal to
- `>` — Greater than
- `<` — Less than
- `>=` — Greater than or equal
- `<=` — Less than or equal

### Logical operators:

- `AND` — Logical AND
- `OR` — Logical OR
- `NOT` — Logical NOT

### Filter modifiers (Builder API):

- `field=value` — Equal
- `field__ne=value` — Not equal
- `field__gt=value` — Greater than
- `field__gte=value` — Greater or equal

- `field__lt=value` — Less than
- `field__lte=value` — Less or equal

### Available Measures

- `degree` — Node degree
- `degree_centrality` — Normalized degree
- `betweenness_centrality` — Betweenness
- `closeness_centrality` — Closeness
- `eigenvector_centrality` — Eigenvector
- `pagerank` — PageRank
- `clustering` — Clustering coefficient

### Working with Results

```
result = Q.nodes().compute("degree").execute(network)

# Iteration
for node in result:
    print(node)

# Count
print(f"Found {result.count} nodes")

# Measures
degrees = result.measures['degree']
for node, degree in degrees.items():
    print(f"{node}: {degree}")

# To pandas
df = result.to_pandas()
```

### I/O API

#### Modern I/O System

```
from py3plex.io import read, write, MultiLayerGraph

# Read (auto-detects format from extension)
graph = read('network.json')
graph = read('network.arrow')
graph = read('network.parquet')

# Write
write(network, 'output.json')
write(network, 'output.arrow')
write(network, 'output.parquet')

# Specify format explicitly
```

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```
graph = read('file.dat', format='json')
write(network, 'file.dat', format='arrow')
```

## Supported Formats

- **JSON** — .json — Human-readable
- **JSONL** — .jsonl — Streaming
- **CSV** — .csv — Spreadsheet-compatible
- **Arrow** — .arrow — High-performance
- **Parquet** — .parquet — Compressed

## Visualization API

### Basic Visualization

```
from py3plex.visualization.multilayer import draw_multilayer_default

# Simple visualization
draw_multilayer_default([network], display=True)

# With layout options
draw_multilayer_default(
    [network],
    display=True,
    layout='force_directed',
    node_size=300,
    edge_width=2
)
```

### Matrix Visualization

```
# Visualize supra-adjacency matrix
network.visualize_matrix({"display": True})
```

## Command-Line Interface

### Basic Usage

```
# Show version
python -m py3plex --version

# Run self-test
python -m py3plex selftest

# Analyze network
python -m py3plex analyze network.edgelist

# Community detection
python -m py3plex communities network.edgelist --algorithm louvain
```

### Summary

This appendix provided quick references for:

- Core API (creating networks, adding nodes/edges)
- Algorithms API (statistics, communities, centrality, dynamics)
- DSL syntax (both Builder API and string syntax)
- I/O API (reading/writing various formats)
- Visualization API
- Command-line interface

#### For detailed documentation:

- Main text chapters for concepts and examples
- Appendix A for repository layout
- Appendix D for error handling
- Online API docs: <https://skblaz.github.io/py3plex/>

### 1.7.23 Bibliography

This section provides references for further reading on multilayer networks and py3plex.

#### Core References

#### Py3plex Publications

#### Community Detection

#### Network Embeddings

#### Dynamics and Processes

#### Applications

#### Biological Networks

#### Social Networks

#### Transportation Networks

#### Software and Tools

#### Online Resources

#### Py3plex:

- Documentation: <https://skblaz.github.io/py3plex/>
- GitHub: <https://github.com/SkBlaz/py3plex>
- PyPI: <https://pypi.org/project/py3plex/>

#### Community:

- Network Science community: <http://www.netscisociety.net/>
- Complex Networks: <https://www.complexnetworks.org/>

#### Datasets:

- Network Repository: <http://networkrepository.com/>
- SNAP: Stanford Network Analysis Project: <http://snap.stanford.edu/data/>
- Netzschleuder: <https://networks.skewed.de/>

## Recommended Reading Path

### For Graduate Students

1. Start with [Kivelä2014] for comprehensive foundations
2. Read [Skrlj2019a] for py3plex overview
3. Read [Mucha2010] for community detection
4. Explore [Bianconi2018] textbook for deeper theory

### For Practitioners

1. Start with [Skrlj2019a] for py3plex capabilities
2. Skim [Kivelä2014] for key concepts
3. Focus on application papers relevant to your domain
4. Use this book and online docs for implementation

### For Researchers

1. Read [Kivelä2014] and [Boccaletti2014] thoroughly
2. Study [Domenico2013] for mathematical rigor
3. Read domain-specific application papers
4. Explore [Bianconi2018] for advanced topics

## 1.7.24 Citation Information

### How to Cite Py3plex

If you use py3plex in your research, please cite the following paper:

#### BibTeX

```
@Article{Skrlj2019,
  author={Skrlj, Blaz and Kralj, Jan and Lavrac, Nada},
  title={Py3plex toolkit for visualization and analysis of multilayer networks},
  journal={Applied Network Science},
  year={2019},
  volume={4},
  number={1},
  pages={94},
  doi={10.1007/s41109-019-0203-7},
  url={https://doi.org/10.1007/s41109-019-0203-7}
}
```

For conference proceedings, you may also cite:

```
@InProceedings{Škrlj2019b,
  author="{\v{S}}krlj, Bla{\v{z}}
  and Kralj, Jan
  and Lavra{\v{c}}, Nada",
  editor="Aiello, Luca Maria
  and Cherifi, Chantal
  and Cherifi, Hocine
  and Lambiotte, Renaud
  and Li{\o}, Pietro
  and Rocha, Luis M.",
  title="Py3plex: A Library for Scalable Multilayer Network Analysis and Visualization",
  booktitle="Complex Networks and Their Applications VII",
  year="2019",
  publisher="Springer International Publishing",
  address="Cham",
  pages="757--768",
  isbn="978-3-030-05411-3",
  doi="10.1007/978-3-030-05411-3_60"
}
```

### APA Style

Škrlj, B., Kralj, J., & Lavrač, N. (2019). Py3plex toolkit for visualization and analysis of multilayer networks. *Applied Network Science*, 4(1), 94. <https://doi.org/10.1007/s41109-019-0203-7>

### MLA Style

Škrlj, Blaz, Jan Kralj, and Nada Lavrač. "Py3plex toolkit for visualization and analysis of multilayer networks." *Applied Network Science* 4.1 (2019): 94.

### Chicago Style

Škrlj, Blaz, Jan Kralj, and Nada Lavrač. "Py3plex toolkit for visualization and analysis of multilayer networks." *Applied Network Science* 4, no. 1 (2019): 94.

### Citing This Book

If you specifically reference this book (rather than the software), please cite:

```
Škrlj, B. (2025). *Practical Multilayer Network Analysis with Py3plex* (Version 1.0).
Available at: https://github.com/SkBlaz/py3plex
```

### BibTeX for This Book

```
@book{Škrlj2025book,
  author = {Škrlj, Blaž},
  title = {Practical Multilayer Network Analysis with Py3plex},
  year = {2025},
  version = {1.0},
  url = {https://github.com/SkBlaz/py3plex}
}
```

## Citing Specific Algorithms

If you use specific algorithms implemented in py3plex, consider citing the original papers:

### Louvain Community Detection

```
@article{Blondel2008,
  title={Fast unfolding of communities in large networks},
  author={Blondel, Vincent D and Guillaume, Jean-Loup and Lambiotte, Renaud and Lefebvre,
↪ Etienne},
  journal={Journal of statistical mechanics: theory and experiment},
  volume={2008},
  number={10},
  pages={P10008},
  year={2008}
}
```

### Multilayer Modularity

```
@article{Mucha2010,
  title={Community structure in time-dependent, multiscale, and multiplex networks},
  author={Mucha, Peter J and Richardson, Thomas and Macon, Kevin and Porter, Mason A and
↪ Onnela, Jukka-Pekka},
  journal={Science},
  volume={328},
  number={5980},
  pages={876--878},
  year={2010}
}
```

### Node2Vec

```
@inproceedings{Grover2016,
  title={node2vec: Scalable feature learning for networks},
  author={Grover, Aditya and Leskovec, Jure},
  booktitle={Proceedings of the 22nd ACM SIGKDD international conference on Knowledge
↪ discovery and data mining},
  pages={855--864},
  year={2016}
}
```

## Software Version Information

Reproducibility best practice is to cite the specific version of py3plex used:

```
import py3plex
print(py3plex.__version__) # e.g., "1.0.2"
```

In your paper's methods section, include:

```
All analyses were performed using py3plex version 1.0.2 [Skr1j2019]
running on Python 3.10.
```

Or in acknowledgments:

This research utilized py3plex (version 1.0.2, Škrlj et al., 2019) for multilayer network analysis.

### License

#### Py3plex Core

The main py3plex library is released under the **MIT License**, a permissive open-source license that allows commercial use:

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#### Bundled Code

**Important:** Some bundled code (e.g., Infomap community detection in `py3plex/algorithms/community_detection/infomap/`) is licensed under **AGPLv3**, which has copyleft requirements.

If you use these features, your application may be subject to AGPLv3 requirements. See Chapter 4 for details on license considerations.

#### This Book

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- **Attribution** — You must give appropriate credit and provide a link to the license

## Acknowledgments

Py3plex development has been supported by:

- Slovenian Research Agency (research program P2-0103 and projects)
- Jožef Stefan Institute
- Contributors and users worldwide

The py3plex community has provided valuable feedback, bug reports, and feature suggestions. We thank all contributors.

Special thanks to:

- **NetworkX developers** — py3plex builds on NetworkX
- **Apache Arrow project** — For high-performance I/O
- **Open-source community** — For tools and libraries

## Contact

For questions about citing py3plex:

- **GitHub Issues:** <https://github.com/SkBlaz/py3plex/issues>
- **Email:** See contact information in the repository

For research collaborations or academic inquiries, see the py3plex repository for current contact information.





## INDICES AND TABLES

- `genindex`
- `modindex`
- `search`



## BIBLIOGRAPHY

- [Kivelä2014] Kivelä, M., Arenas, A., Barthelemy, M., Gleeson, J. P., Moreno, Y., & Porter, M. A. (2014). Multilayer networks. *Journal of Complex Networks*, 2(3), 203-271. <https://doi.org/10.1093/comnet/cnu016>  
Comprehensive review of multilayer network theory, mathematics, and applications. Essential reading.
- [Boccaletti2014] Boccaletti, S., Bianconi, G., Criado, R., Del Genio, C. I., Gómez-Gardenes, J., Romance, M., ... & Zanin, M. (2014). The structure and dynamics of multilayer networks. *Physics Reports*, 544(1), 1-122. <https://doi.org/10.1016/j.physrep.2014.07.001>  
Physics-focused perspective on multilayer networks with emphasis on dynamics.
- [Bianconi2018] Bianconi, G. (2018). *Multilayer Networks: Structure and Function*. Oxford University Press.  
Comprehensive textbook covering theory, methods, and applications.
- [Domenico2013] De Domenico, M., Solé-Ribalta, A., Cozzo, E., Kivelä, M., Moreno, Y., Porter, M. A., ... & Arenas, A. (2013). Mathematical formulation of multilayer networks. *Physical Review X*, 3(4), 041022. <https://doi.org/10.1103/PhysRevX.3.041022>  
Rigorous mathematical formulation of multilayer network representations.
- [Škrlić2019a] Škrlić, B., Kralj, J., & Lavrač, N. (2019). Py3plex toolkit for visualization and analysis of multilayer networks. *Applied Network Science*, 4(1), 94. <https://doi.org/10.1007/s41109-019-0203-7>  
Primary py3plex paper. Cite this when using py3plex in research.
- [Škrlić2019b] Škrlić, B., Kralj, J., & Lavrač, N. (2019). Py3plex: A library for scalable multilayer network analysis and visualization. In *International Conference on Complex Networks and Their Applications* (pp. 757-768). Springer. [https://doi.org/10.1007/978-3-030-05411-3\\_60](https://doi.org/10.1007/978-3-030-05411-3_60)  
Conference paper focusing on visualization and scalability.
- [Blondel2008] Blondel, V. D., Guillaume, J. L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2008(10), P10008. <https://doi.org/10.1088/1742-5468/2008/10/P10008>  
Louvain algorithm for community detection.
- [Rosvall2008] Rosvall, M., & Bergstrom, C. T. (2008). Maps of random walks on complex networks reveal community structure. *Proceedings of the National Academy of Sciences*, 105(4), 1118-1123. <https://doi.org/10.1073/pnas.0706851105>  
Infomap algorithm for community detection.
- [Mucha2010] Mucha, P. J., Richardson, T., Macon, K., Porter, M. A., & Onnela, J. P. (2010). Community structure in time-dependent, multiscale, and multiplex networks. *Science*, 328(5980), 876-878. <https://doi.org/10.1126/science.1184819>

Multilayer community detection with modularity optimization.

- [Grover2016] Grover, A., & Leskovec, J. (2016). node2vec: Scalable feature learning for networks. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 855-864). <https://doi.org/10.1145/2939672.2939754>

Node2Vec algorithm for network embeddings.

- [Perozzi2014] Perozzi, B., Al-Rfou, R., & Skiena, S. (2014). DeepWalk: Online learning of social representations. In *Proceedings of the 20th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 701-710). <https://doi.org/10.1145/2623330.2623732>

DeepWalk algorithm for learning node representations.

- [Pastor-Satorras2015] Pastor-Satorras, R., Castellano, C., Van Mieghem, P., & Vespignani, A. (2015). Epidemic processes in complex networks. *Reviews of Modern Physics*, 87(3), 925. <https://doi.org/10.1103/RevModPhys.87.925>

Comprehensive review of epidemic modeling on networks.

- [Buldyrev2010] Buldyrev, S. V., Parshani, R., Paul, G., Stanley, H. E., & Havlin, S. (2010). Catastrophic cascade of failures in interdependent networks. *Nature*, 464(7291), 1025-1028. <https://doi.org/10.1038/nature08932>

Cascading failures in interdependent networks.

- [Gligorijevic2019] Gligorijević, V., Barot, M., & Bonneau, R. (2019). deepNF: deep network fusion for protein function prediction. *Bioinformatics*, 35(5), 757-766. <https://doi.org/10.1093/bioinformatics/bty440>

Application to biological network analysis.

- [Magnani2013] Magnani, M., & Rossi, L. (2013). Pareto distance for multilayer network analysis. In *Social Computing, Behavioral-Cultural Modeling and Prediction* (pp. 249-256). Springer. [https://doi.org/10.1007/978-3-642-37210-0\\_27](https://doi.org/10.1007/978-3-642-37210-0_27)

Social network analysis with multilayer methods.

- [Gallotti2014] Gallotti, R., & Barthelemy, M. (2014). Anatomy and efficiency of urban multimodal mobility. *Scientific Reports*, 4(1), 6911. <https://doi.org/10.1038/srep06911>

Transportation networks as multilayer systems.

- [NetworkX] Hagberg, A., Swart, P., & S Chult, D. (2008). Exploring network structure, dynamics, and function using NetworkX. *Los Alamos National Lab.(LANL), Los Alamos, NM (United States)*.

NetworkX documentation: <https://networkx.org/>

- [Arrow] Apache Arrow Development Team. (2023). Apache Arrow. <https://arrow.apache.org/>

High-performance columnar data format.

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